Redrawn Boundaries, Unequal Campaigns: Constituency Economic Context and Support for Reform UK.

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Abstract

Constituencies organize the political competition through which economic geography can shape radical right support. I examine how assignment to a constituency of lower economic standing affects support for Reform UK during the 2024 UK general election. The 2023 boundary review in England created differences in constituency income deprivation among British Election Study respondents from the same old constituency with stable observed geographic assignment. Under conditional parallel change, this comparison identifies the effect of constituency economic context. Assignment to a more income deprived constituency raises Reform support by 2.5 percentage points per standard deviation of deprivation across old constituencies. Conditional on old constituency, Reform's expenditure per elector is higher relative to combined Conservative and Liberal Democrat expenditure in more deprived destinations, with historical electoral opportunity accounting for part of the latter parties' gradients. These findings identify a total contribution of constituency economic context to radical right support, while the transmission from resources and communication to voters remains unresolved.

Introduction

The economic geography of radical right support raises a question about how economic conditions become politically consequential. Research on European discontent links support for radical right and anti EU parties to regional economic decline and uneven development (Rodríguez-Pose, 2018; Dijkstra et al., 2020; Rodríguez-Pose et al., 2023). These patterns connect economic trajectories to the political choices available in different areas. I examine how the economic standing of the constituency in which voters cast their ballots affects support for Reform UK.

Constituencies organize electoral competition around geographically defined electorates. In Britain, parties direct campaign resources towards seats where they expect electoral returns (Denver et al., 2003; Fisher, Fieldhouse, et al., 2016). The constituency therefore connects an electorate's economic composition to the candidates, campaigning and competitive opportunities its voters encounter. Its economic standing can acquire political significance through this organized electoral environment.

I argue that constituency economic context shapes radical right support through the interaction of political supply and contextual demand. Parties allocate attention and resources across constituencies, while voters interpret their circumstances through the competition and communication around them. A constituency of lower economic standing may offer Reform a more favourable environment because of the relative presence of competing parties, the credibility of appeals about neglect, or their interaction. Such an environment can matter to voters in different personal economic positions, making constituency context relevant to the political consequences of economic geography.

Estimating this contextual effect requires separating changes in the electorate around voters from changes in their own circumstances and local economies. Trade shocks and welfare cuts can change both together (Colantone and Stanig, 2018; Fetzer, 2019), while residential selection contributes to differences between areas (Gallego et al., 2016). Before

the 2024 UK general election, the Boundary Commission for England redrew parliamentary constituencies to equalize their electorates under published statutory criteria (Boundary Commission for England, 2023b). This boundary review divided many old constituencies among new constituencies of different economic standing. It created a comparison between voters who shared an old constituency but were assigned to different electoral environments.

I follow respondents in the British Election Study Internet Panel through the 2024 campaign using its combined release through wave 30 (The British Election Study Team, 2025). The primary comparison uses waves 26 and 29, before the formal campaign and after the election, among respondents with stable observed geographic assignment. Wave 26 follows the map's legal establishment, so the comparison concerns subsequent changes in support. I apply the same frozen income deprivation measure to both constituency systems and compare changes among respondents from the same old constituency. Under conditional parallel change, this design identifies the effect of assignment to a constituency of different economic standing during the observed campaign comparison.

My results show that assignment to a more income deprived constituency raised Reform support over this comparison. An increase of 1 standard deviation in income deprivation across old constituencies corresponds to 2.5 percentage points of additional support, against a pre campaign level of 15%. This exposure unit is 5.31 percentage points of the deprivation rate, rather than the standard deviation of respondents' actual shifts. The political allocation evidence shows lower Conservative and Liberal Democrat candidate expenditure in more deprived destinations. On a matched respondent sample, Reform's expenditure per elector is higher relative to combined Conservative and Liberal Democrat expenditure in more deprived destinations, conditional on old constituency, although its absolute spending gradient is imprecise. This directly estimated relative allocation pattern describes the political environment accompanying the support effect.

The contribution is causal evidence that constituency economic context shapes radical right support under conditional parallel change. The allocation evidence locates an observable

political difference within that context. Political supply and contextual demand can work together within the identified total contextual effect. The case shows how district organized competition can give economic geography political consequences across personal economic positions.

Theory

Constituency economic context connects the composition of an electorate to the political environment organized around it. The constituency defines a group of voters for whom parties select candidates, allocate effort and develop appeals. Its economic standing can therefore matter through both the circumstances of its residents and the competition directed towards them. I use constituency context to describe this combined electoral environment, whose composition and political organization may shape support for Reform UK.

Economic insecurity and social status provide reasons for expecting economic geography to matter politically. Research connects radical right support to household unemployment risk, declining occupational positions, status anxiety and positional deprivation (Abou-Chadi and Kurer, 2021; Gidron and Hall, 2017; Kurer, 2020; Engler and Weisstanner, 2021; Burgoon et al., 2019). These accounts link material circumstances to how people understand their position relative to others. A constituency's economic composition may make some concerns more politically salient or credible even among residents whose own finances are secure. This possibility places contextual demand within the argument alongside the organization of electoral competition.

Electoral districts give parties a unit in which to organize that competition. The radical right's electoral prospects depend partly on its organization and message, mainstream strategy and electoral institutions (Mudde, 2007; Alonso and Rovira Kaltwasser, 2015). In Britain, party headquarters coordinate constituency campaigns and concentrate resources in strategically important seats (Denver et al., 2003; Fisher, Fieldhouse, et al., 2016). Constituency economic standing may be relevant to these decisions because it describes the electorate among whom parties assess their electoral opportunities. That connection makes allocation a potential component of the contextual effect, with the specific role of economic standing requiring empirical examination.

Strategic targeting can create different political environments for voters in different

constituencies. Parties direct resources where they expect electoral dividends, and the effectiveness of campaigning varies with electoral context (Fieldhouse et al., 2019; Fisher, Cutts, et al., 2018; C. J. Pattie and Cutts, 2025). An electorate's economic composition may overlap with historical partisan strength and the opportunities parties perceive there. The resulting allocation can favour one party relative to particular competitors even when its own expenditure is similar across seats. The relevant comparison is therefore party specific and depends on whether resources are measured in absolute or relative terms.

Voters encounter these differences as they decide whether to maintain an existing preference or support another party. Research on constituency campaigning links electoral effects to tactical position and incumbency, while work on campaign contact connects it to loyalty and defection (C. Pattie, Hartman, et al., 2017; Núñez, 2025). This gives campaigning potential relevance to both the acquisition and persistence of radical right support. Contact recalled in surveys is itself conditioned by partisanship, which complicates inference about what voters actually received (C. Pattie, Whitworth, et al., 2015). An observed electoral transition consequently describes a response pattern within the constituency environment, with its causal connection to particular campaign activities requiring separate evidence.

Communication and social interpretation can reinforce these competitive processes. Accounts of neglect and community decline connect collective conditions to political judgement (Cramer, 2016; Munis, 2022; Kenny and Luca, 2021), while national rhetoric can politicize local conditions (Hopkins, 2010; Hopkins, 2011). Residents of low income communities also report more negative assessments of their community's representation (McKay, 2019). A constituency of lower economic standing may make claims about neglected communities more relevant or credible. Reform can connect those claims to a challenge against established parties, giving dissatisfaction with representation a radical right electoral outlet. The appeal can reach financially secure residents who judge the constituency's treatment adversely, as well as those experiencing personal insecurity. Its electoral force may also depend on the relative effort established competitors devote to the seat. Campaigns and contextual demand

can therefore reinforce each other, whether or not voters recognize that their constituency assignment has changed.

Research on other contextual changes illustrates how political interpretation can accompany an estimated electoral effect. Alrababah and colleagues estimate increased anti immigrant party support following Swiss border liberalization and document overcrowding narratives directed towards exposed municipalities (Alrababah et al., 2025). Their evidence on political awareness is suggestive, with the causal role of narrative supply left open. Dickson and colleagues estimate that exposure to GP practice closures increased radical right support in two British panels and document party communications linking NHS decline to immigration (Dickson et al., 2026). Their framing account interprets the total exposure effect and its variation, rather than estimating persuasion by the messages themselves. These studies provide substantive foundations for examining how political communication and local conditions work together.

My core expectation is that voters assigned to a more income deprived constituency become more likely to support Reform. The expectation concerns economic standing as a characteristic of the electoral environment, measured with frozen economic inputs. Party allocation, perceived electoral options and communication offer different observable traces of how that environment is organized. Support among personally secure voters would establish the reach of the contextual relationship, while variation across personal circumstances remains an empirical question. Specific allocation gradients and transitions are exploratory implications of this argument, rather than a set of retrospectively confirmed predictions.

The argument applies most directly where parties organize electoral competition around districts. Voters can enter an already differentiated electoral environment through reassignment, so its political consequences can emerge as an election approaches without a new deterioration in local economic conditions or a party response to the redrawing itself. The broader account developed from the contextual finding and the subsequent investigation of political organization.

Research Design

The 2023 boundary review changed the constituency electorate around voters, creating variation in economic context within old constituencies. The Boundary Commission for England conducted the review under the Parliamentary Constituencies Act 1986 as amended in 2020 (Johnston and Uberoi, 2023; Boundary Commission for England, 2023b). England received 543 of the UK's 650 seats. Except for the two Isle of Wight constituencies, English seats had to contain between 69,724 and 77,062 electors, within 5% of the electoral quota of 73,393 (Boundary Commission for England, 2023a; Boundary Commission for England, 2023b).

Within that range, statutory considerations included geography, local government and existing constituency boundaries, local ties and the inconvenience of change. The Commission's Guide excludes existing voting patterns and prospective party fortunes, while socioeconomic evidence enters principally through local ties (Boundary Commission for England, 2023a; Boundary Commission for England, 2023b). These rules constrain the redrawing but can preserve local compositional differences between areas assigned to different successors. The design therefore uses comparisons within old constituencies and evaluates parallel change, rather than treating the deprivation shift as randomized.

After public consultations, the Parliamentary Constituencies Order 2023 was made on 15 November 2023 and came into force on 29 November 2023 (UK Government, 2023). The first UK general election under the new map took place on 4 July 2024 (Barr et al., 2024). Only 55 of England's 533 old constituencies remained completely unchanged (Boundary Commission for England, 2023b). Of the old constituencies, 418 were divided among multiple successors. Figure 1 maps the average standardized difference between respondents' new and old constituency income deprivation, showing the geographic distribution of these shifts.

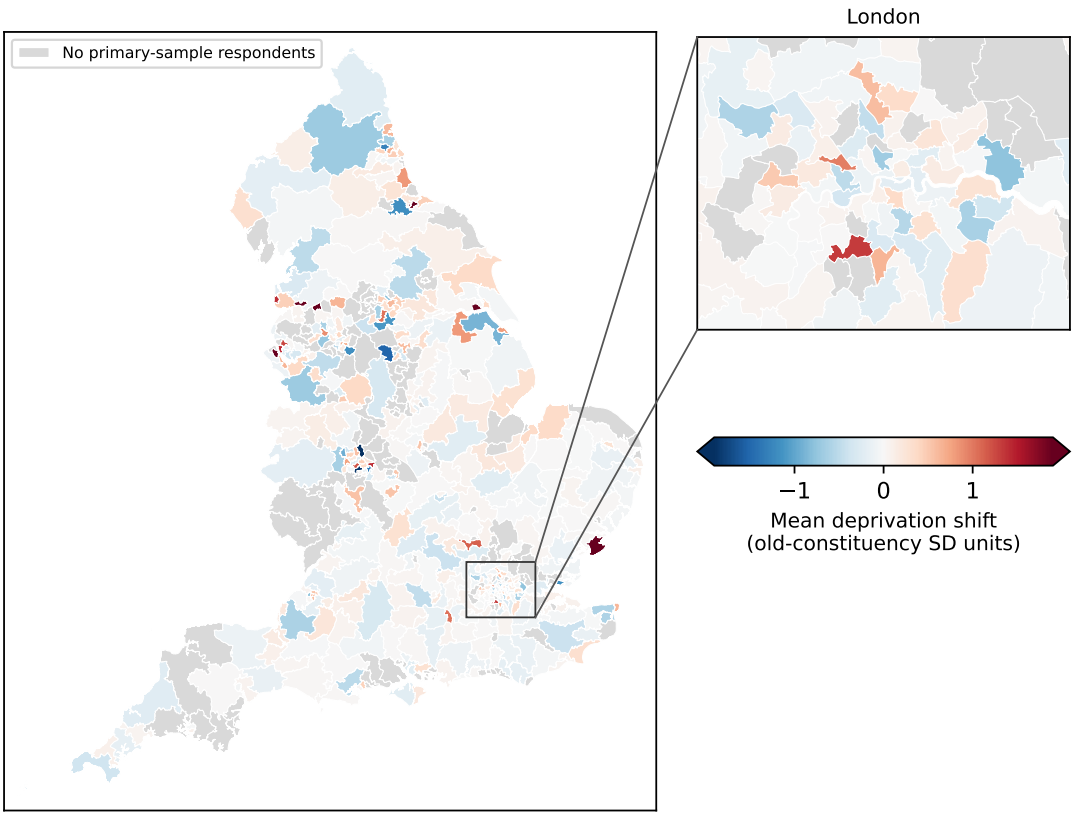


Figure 1: Mean deprivation shift by 2024 constituency in England. Contains ONS and OS data © Crown copyright and database right 2024

I link respondents' observed constituency assignments to their support before and during the campaign using the British Election Study Internet Panel's combined release through wave 30 (The British Election Study Team, 2025). Wave 26 ran from 3 to 22 May 2024, after the map became law but before the formal campaign. Waves 27 and 28 ran from 24 May to 7 June and 10 June to 3 July, and wave 29 from 5 to 19 July. The primary outcome is the change from Reform UK vote intention at W26 to reported vote at W29. The W26 to W28 endpoint retains vote intention wording to assess support during the campaign on a consistent measure. Wave 30, fielded in May 2025, supplies a later follow up rather than a 2024 campaign observation.

The primary sample contains 7,685 respondents from 334 old and 418 new constituencies, linked through 764 overlap pairs. Respondents have valid W26 and W29 outcomes, began

in an old constituency divided among successors, and retain the same recorded old and new constituency codes across observed waves W19 to W29. This restriction establishes stable observed geographic assignment, with residential mobility inside those units remaining unobserved. The original sample of 8,766 required matching codes at W26 and W29. Table 1 retains its 2.18 percentage point estimate alongside the current sample's 2.51 point estimate per old constituency standard deviation of income deprivation. The longer stability rule was defined before its estimate was inspected, but its promotion to primary occurred after the original result and the restricted sample result were known. The measurement rationale is to align the dose and earlier responses with the observed constituency assignment, and the full deviation history is retained in the preanalysis appendix.

Economic standing is measured by the income domain of the English Indices of Deprivation 2019, the proportion of residents experiencing low income deprivation, including people in work on low earnings (Ministry of Housing, Communities and Local Government, 2019a). The underlying benefit and tax credit information refers to 2015/16 (Ministry of Housing, Communities and Local Government, 2019b). Applying this frozen measure to both maps isolates differences in constituency composition generated by the redrawing from economic change in the inputs. Employment deprivation and the overall Index of Multiple Deprivation provide alternative measures.

I aggregate the 32,844 lower layer super output areas (LSOAs) using 2015 population weights (Ministry of Housing, Communities and Local Government, 2019a; Ministry of Housing, Communities and Local Government, 2019b). Population weighted centroids assign 2011 LSOAs to the December 2019 constituency map, and the official best fit lookup bridges to 2021 LSOAs and the July 2024 map. I check the constituency pairs against the official Commons overlap workbook (House of Commons Library, 2024). Appendix Table A1 compares the constructed rates with the independent Commons deprivation rankings (Francis-Devine, 2019). The income deprivation ranking closely agrees with the constructed old constituency rates. The standard deviation of income deprivation across old constituencies is

5.31 percentage points, which supplies the exposure unit throughout.

The dose is the signed difference in constituency income deprivation between the new and old maps, standardized by the old constituency cross sectional standard deviation,

$$D_i = \frac{Z_{c_{2024}(i)} - Z_{c_{2019}(i)}}{\sigma_{2019}}, \tag{1}$$

where Z_c is the frozen income deprivation rate and $c_{2019}(i)$ and $c_{2024}(i)$ are respondent i's old and new constituencies. A positive dose denotes assignment to a more deprived constituency. Respondents sharing an old and new constituency pair receive the same dose, whatever their personal income. I estimate

$$Y_{i,29} - Y_{i,26} = \alpha_{g(i)} + \beta D_i + \varepsilon_i, \tag{2}$$

where $g(i)$ denotes the old constituency and Y indicates Reform support. First differencing removes stable respondent differences and is equivalent to respondent fixed effects with two waves (Schmidheiny and Siegloch, 2023). Old constituency fixed effects restrict the comparison to respondents sharing an origin but receiving different doses. Standard errors are clustered two ways by old origin and new destination, allowing correlated responses within either constituency.

Conditional parallel change requires that these respondents would have changed their support similarly had the boundaries stayed in place. First differencing accommodates stable local differences in support levels, while differential local political change remains a threat. A swing towards Reform in more deprived parts of an old constituency that coincided with the campaign could generate a dose gradient even under unchanged boundaries. Earlier support trajectories, predetermined covariate adjustment and gradient sensitivities assess this assumption and its vulnerabilities.

The coefficient summarizes the total contextual effect, bundling composition, candidates,

campaign effort, competitive position and communication. Separating these components requires additional assumptions about mediators (Acharya et al., 2016; Fu and Slough, 2026). Campaigns and conversations can cross constituency lines, so the comparison holds neighbouring assignments at their realized values rather than moving everyone simultaneously (Hudgens and Halloran, 2008; Aronow and Samii, 2017). The linear dose response also summarizes effects across different shifts with regression weights (Callaway et al., 2024; Chaisemartin and D'Haultfœuille, 2020). Appendix Table A17 reports those weights alongside the binned relationship in Appendix Figure A9.

To examine the timing of support changes, I estimate

$$Y_{it} = \mu_i + \lambda_{g(i),t} + \sum_{s \neq 26} \beta_s D_i \mathbb{1}(t = s) + \varepsilon_{it}, \tag{3}$$

where μ_i are respondent fixed effects and $\lambda_{g(i),t}$ are old constituency by wave fixed effects. The reference is W26, before the formal campaign but after the map's legal establishment. Coefficients for W19 to W25 assess earlier divergence, W27 to W29 trace the campaign comparison, and W30 assesses later support. Pointwise intervals accompany simultaneous confidence bands covering the wave coefficients jointly.

Design Validation

The contextual comparison rests on parallel change among respondents from the same old constituency who received different deprivation shifts. First differencing removes stable differences in support levels. Earlier differential trends and a local swing specific to 2024 remain the central identifying threats.

Figure 2 plots coefficients on the deprivation shift from the first panel wave in 2014 relative to W26, with 95% confidence intervals. The unit is percentage points of radical right support per old constituency standard deviation of income deprivation, with two way old and new

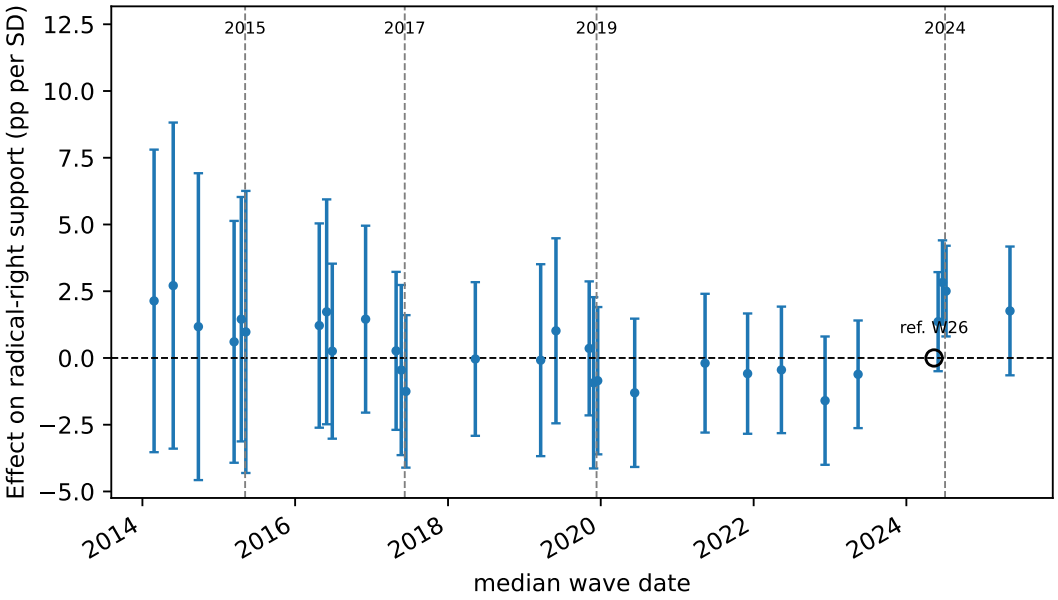


Figure 2: Radical right support and destination disadvantage, W1 to W30

constituency clustering unless stated otherwise. The 25 coefficients before the campaign are jointly indistinguishable from zero (Wald $\chi^2 = 14.54$, $df = 25$, $p = 0.951$, $N = 7{,}685$). The equivalence diagnostic asks how small all earlier differences can jointly be judged to be, using the largest bound across the pre period coefficients. That bound exceeds the estimated effect on mean W28 and W29 support, so the data fail to establish that all earlier differences were small relative to that effect. Together these diagnostics provide favourable evidence on earlier trajectories while leaving uncertainty about parallel change.

Panel A of Figure 3 applies the same dose to changes in radical right support around earlier general elections, using the last wave before and first wave after polling day. Appendix Table A45 reports the estimates for respondents with matching old constituency codes across each comparison. The estimates are −1.00 percentage points in 2015 (95% CI −3.63 to 1.63, $N = 1{,}085$), −0.53 in 2017 (−3.55 to 2.49, $N = 1{,}915$) and −0.81 in 2019 (−2.63 to 1.01, $N = 2{,}918$). Their direction differs from the 2024 estimate, although the wide intervals permit substantial uncertainty.

Figure 4 displays joint Wald statistics for predetermined covariates and identifying

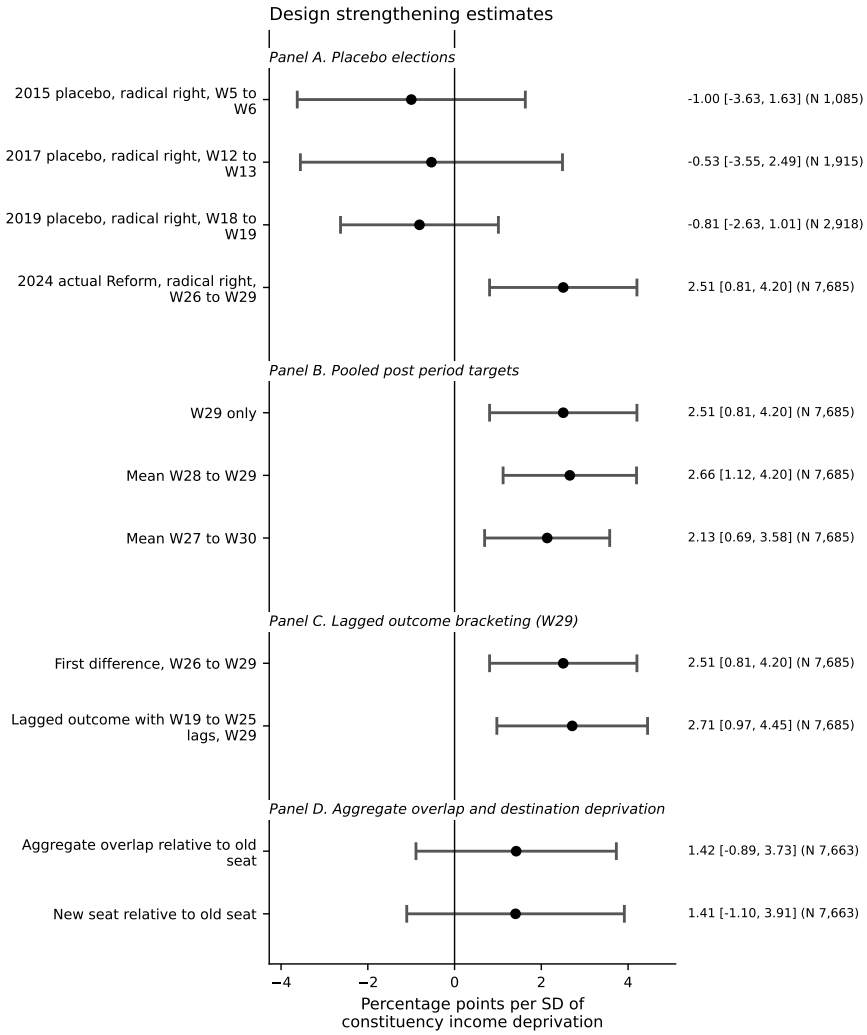


Figure 3: Additional tests of the design

diagnostics, with full estimates in Appendix Table A46. The W25 audit rejects exact balance across 18 predetermined characteristics (Wald $\chi^2 = 46.14$, $df = 18$, $p < 0.001$), with household income the largest individual difference. The design therefore requires parallel change conditional on old constituency, rather than balanced assignment. Table 1 reports 2.35 percentage points after full predetermined adjustment (95% CI 0.69 to 4.01, $N = 7{,}685$). Covariate by wave adjustment also allows referendum vote, 2019 vote, age, education, tenure and gender to predict different changes in support. Lagged outcome adjustment (Angrist and Pischke, 2009) and linear trends on calendar time and survey wave scales provide further

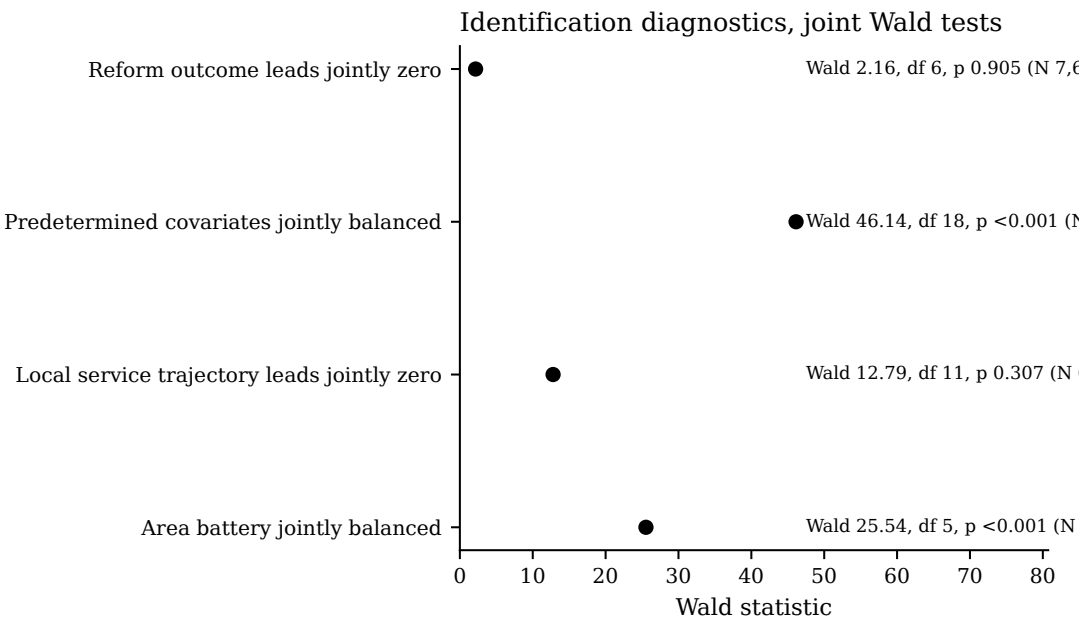


Figure 4: Joint identification diagnostics

checks.

A coincident swing towards Reform in more deprived areas requires a separate sensitivity analysis. I measure part deprivation as frozen aggregate old and new constituency overlap income deprivation, divided by the cross sectional old constituency income deprivation standard deviation. Respondents sharing a code pair share this aggregate exposure. The auxiliary comparison measures W26 to mean W28 and W29 support change among 11,799 respondents across England, including unchanged constituencies. It pairs a dose estimate with old constituency by wave fixed effects with a national deprivation gradient estimated using region by wave fixed effects. Within old constituencies, larger doses accompany more deprived overlap areas. I denote this relationship by δ and the national support change per unit of this normalized overlap measure by γ. The quantity $\gamma\delta$ represents the portion of the auxiliary pooled dose estimate that such a national swing could account for.

Figure 5 plots the auxiliary pooled estimate after subtracting this quantity, with a 95% bootstrap band clustered by old constituency. The netted estimate is in percentage points

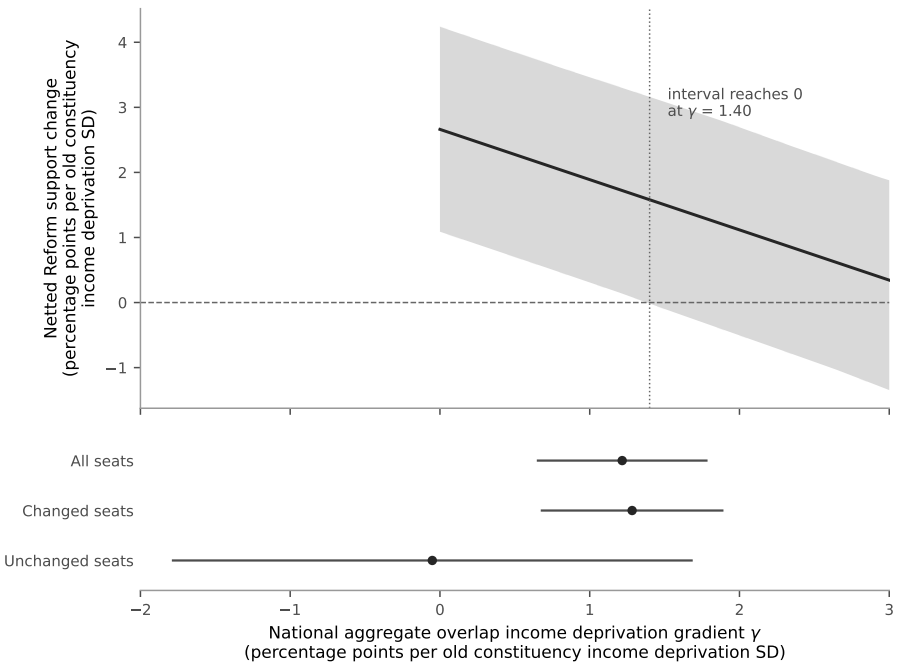


Figure 5: Reassignment effect net of a national part deprivation gradient

per old constituency income deprivation standard deviation of the reassignment dose. On the saved grid of gradients in 0.05 increments, $\gamma = 1.40$ is the first point whose band includes zero. The estimated national gradient is 1.22 percentage points in these units (95% CI 0.65 to 1.78, $N = 11{,}799$). At that gradient the netted estimate is 1.72 points (old constituency cluster bootstrap 95% CI 0.15 to 3.36). The national gradient has divergent prior trajectories (joint lead test, Wald $\chi^2 = 23.63$, $df = 6$, $p < 0.001$), making it a sensitivity benchmark rather than an identified 2024 shock. A differential local swing remains a substantive threat even though the netted interval at this benchmark stays above zero.

Appendix diagnostics extend these checks to departures from parallel change, selection and inference. Honest parallel change bounds assess departures from earlier trends, and entropy balancing reweights the complete case sample on predetermined characteristics (Appendix Table A47). The omitted confounder sensitivity is benchmarked to left right self placement (Appendix Table A23). Table 1 collects the primary and historical samples, the consistent

W26 to W28 endpoint, covariate adjustment and inverse probability of observation weighting. Appendix checks retain anticipation and placebo endpoints, multiple testing correction, influence diagnostics, wild cluster bootstrap inference and alternative exposure and stability restrictions. Weighting remains a sensitivity because the available W19 to W29 longitudinal weight covers different waves from the primary W26 to W29 comparison.

Results

Figure 6 plots how the effect of constituency economic context on Reform UK support develops over time under the parallel change assumption. It shows the coefficient on the deprivation shift at each wave relative to W26, the last wave before the formal campaign, with pointwise and simultaneous 95% confidence intervals. The horizontal axis separates earlier observations from the campaign and subsequent waves. The vertical axis gives percentage points of Reform support per old constituency standard deviation of income deprivation, equivalent to 5.31 percentage points of the deprivation rate. This denominator describes economic differences across old constituencies, rather than the dispersion of respondents' boundary induced shifts. Each comparison holds the old constituency constant.

Table 1 presents the primary endpoint estimate beside the original sample and the principal sensitivity comparisons. Between W26 and W29, assignment to a constituency with income deprivation higher by 1 old constituency standard deviation raises Reform support by 2.51 percentage points (two way clustered SE 0.87, 95% CI 0.81 to 4.20, $p = 0.004$, $N = 7{,}685$). The model compares individual changes among respondents from the same old constituency and clusters standard errors by both old origin and new destination. The current sample requires stable observed geographic assignment over W19 to W29. The original endpoint stability sample yields 2.18 percentage points per old constituency standard deviation (two way clustered SE 0.80, 95% CI 0.62 to 3.75, $p = 0.006$, $N = 8{,}766$). I promoted the longer observed stability restriction after examining outcomes, so the original comparison remains

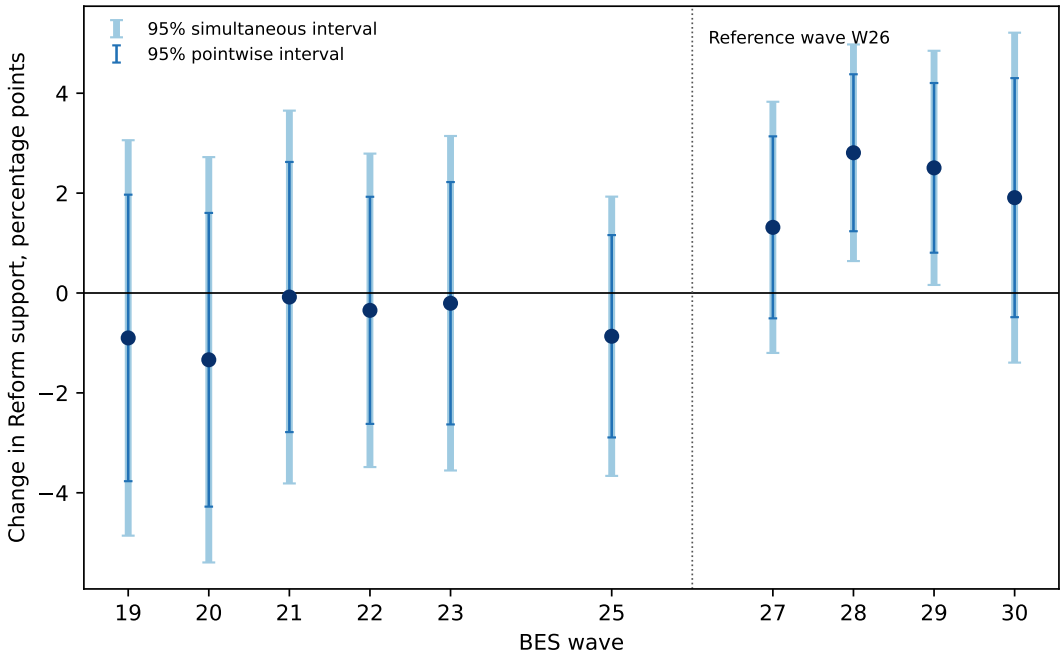


Figure 6: Constituency deprivation and Reform support over time

visible alongside it. Both restrictions concern recorded assignment, rather than verified residential immobility.

Table 1: Primary estimate and validation checks

Specification	Estimate	SE	95% CI	p	N
Primary W26 to W29 estimate	2.51	0.87	[0.81, 4.20]	0.004	7,685
Controlling for reform-induced incumbent loss	2.44	0.85	[0.78, 4.11]	0.004	7,685
Original endpoint stability sample	2.18	0.80	[0.62, 3.75]	0.006	8,766
W26 to W28 intention endpoint	2.34	0.79	[0.80, 3.88]	0.003	8,837
Full predetermined adjustment	2.35	0.85	[0.69, 4.01]	0.006	7,685
Full local perception adjustment	2.39	0.85	[0.72, 4.07]	0.005	7,685
Calendar time trend adjustment	2.49	0.87	[0.78, 4.20]	0.004	7,685
Attrition weighting	2.48	0.92	[0.68, 4.27]	0.007	7,685
More versus less or equally deprived, overlap weighted	2.11	1.06	[0.04, 4.18]	0.046	4,938

The timing in Figure 6 places the divergence during the campaign, subject to the uncertainty in the earlier comparisons. The six leads from W19 to W25 are jointly consistent with zero (two way clustered Wald $\chi^2 = 2.16$, $df = 6$, $p = 0.904536$, $N = 7{,}685$). The simultaneous equivalence diagnostic nevertheless fails to place all six earlier coefficients within

a band as narrow as the primary effect. The favourable joint test therefore leaves substantively important differential earlier change compatible with the uncertainty. At W27, the coefficient is 1.31 percentage points per old constituency standard deviation (two way clustered SE 0.93, 95% CI −0.51 to 3.14, $p = 0.157$, $N = 7{,}685$). By W28 it is 2.81 percentage points in the same exposure unit (two way clustered SE 0.80, 95% CI 1.24 to 4.38, $p < 0.001$, $N = 7{,}685$). The W30 estimate is 1.91 percentage points per old constituency standard deviation (two way clustered SE 1.22, 95% CI −0.48 to 4.30, $p = 0.118$, $N = 7{,}685$), leaving persistence into May 2025 uncertain.

The comparisons in Table 1 also locate the result within the design's measured compositional differences. The predetermined balance audit rejects exact balance, making the covariate adjusted estimates important complements to the primary comparison. Adjustment for the full predetermined covariate set yields 2.35 percentage points per old constituency standard deviation (two way clustered SE 0.85, 95% CI 0.69 to 4.01, $p = 0.006$, $N = 7{,}685$). This stability supports the contextual interpretation conditional on parallel change, while differential local political change correlated with boundary placement remains the principal confounding threat. The appendix retains the influence, wild cluster bootstrap, attrition, placebo, simultaneous band and linear trend sensitivities that assess this interpretation.

Figure 7 compares the main specifications on the same percentage point scale, with horizontal lines marking two way clustered 95% intervals. The measurement consistent W26 to W28 intention comparison yields 2.34 percentage points per old constituency standard deviation (two way clustered SE 0.79, 95% CI 0.80 to 3.88, $p = 0.003$, $N = 8{,}837$). This positive intention endpoint provides corroborating evidence before retrospective vote reporting, although its different sample leaves the contribution of recall to the primary comparison unresolved. Retaining respondents who answered Don't know at baseline yields 2.70 percentage points per old constituency standard deviation for W26 to W29 (two way clustered SE 0.95, 95% CI 0.85 to 4.56, $p = 0.004$, $N = 9{,}207$). The alternative endpoint and baseline coding thus preserve a positive association with constituency economic context.

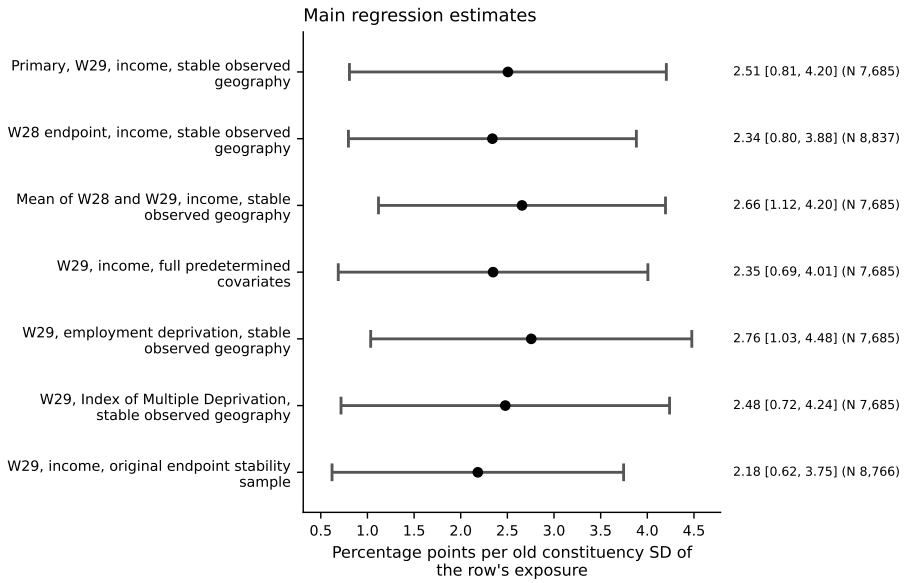


Figure 7: Reform support across the main specifications

The scope of the contextual effect includes voters with secure personal economic circumstances. Appendix Table A28, Panels A and B, compares the estimates across respondents' economic positions. The joint test of six personal economic interactions is imprecise (two way clustered Wald $\chi^2 = 3.36$, $df = 6$, $p = 0.763$, $N = 3{,}986$), leaving differences across positions unresolved. These estimates support a contextual interpretation among personally secure respondents while allowing heterogeneity that the subgroup comparisons estimate only imprecisely. Appendix Figure A20 retains the other party outcomes, and Appendix Table A20 reports Reform support by baseline intention. The primary post election support frame comprises recorded voters, so its nonvoting comparison describes a conditional frame rather than a population turnout effect.

Political Organization

Constituency economic standing accompanies an unequal allocation of political effort that can make Reform UK more competitive relative to established rivals. Figure 8 distinguishes

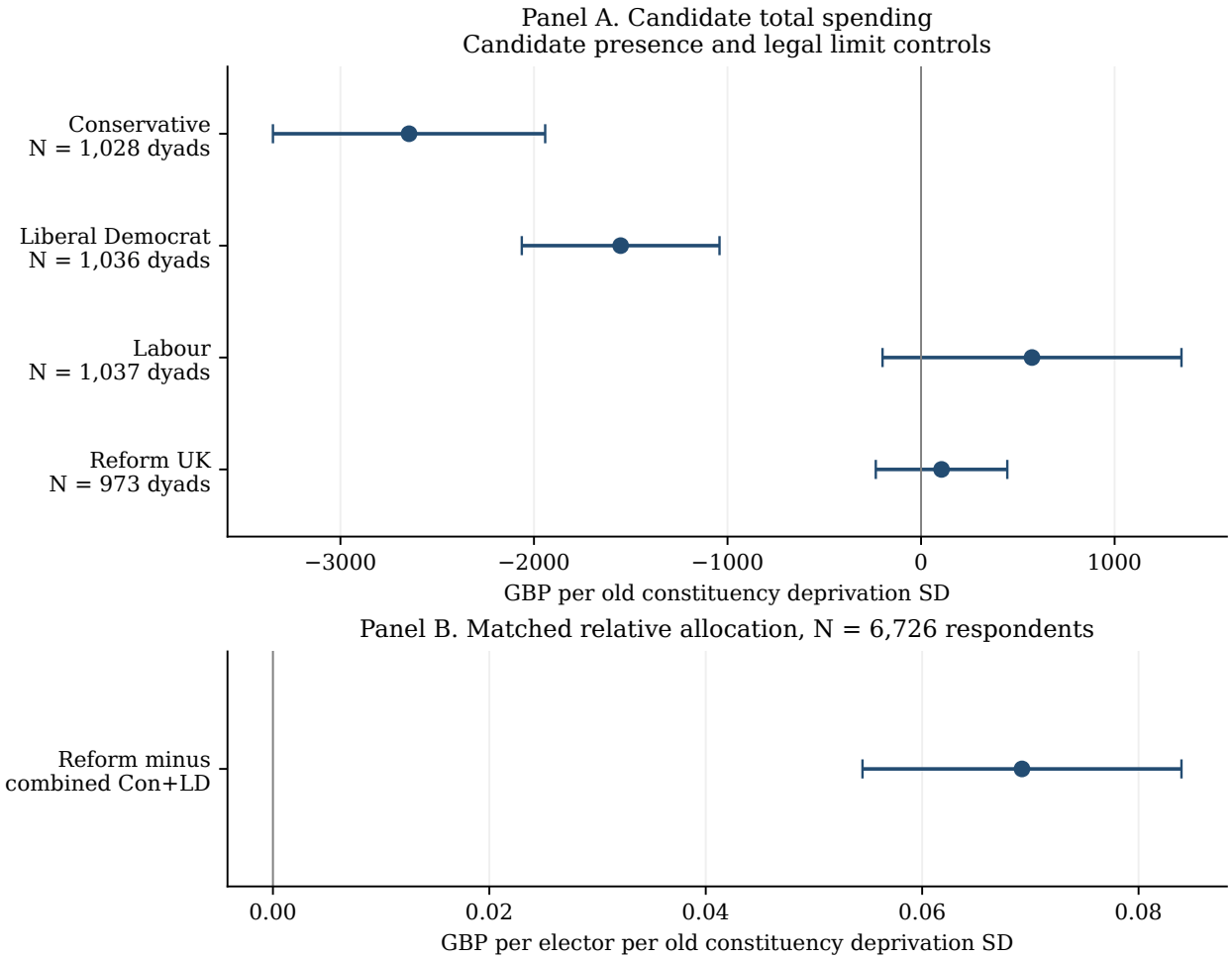


Figure 8: Candidate spending and matched relative allocation

candidate expenditure in pounds from a matched comparison of expenditure per elector experienced by respondents. The matched outcome subtracts combined Conservative and Liberal Democrat expenditure per elector from Reform expenditure per elector on identical respondent rows. Its gradient is £0.06923 per elector per old constituency standard deviation of income deprivation (two way clustered SE 0.00752, 95% CI 0.05449 to 0.08396, $p < 0.001$, Holm adjusted $p < 0.001$, $N = 6{,}726$). Old constituency fixed effects hold origins constant, and paired covariance accounts for the common observations underlying the party comparison. This relative allocation pattern gives affirmative evidence about the political environment accompanying the support effect. It describes a relative expenditure advantage, with the resources to votes connection remaining a separate empirical question.

The candidate level comparisons in Figure 8 identify which parties contribute to this uneven environment. These regressions use old and new constituency overlap dyads, hold old origin, candidate presence and the legal spending limit constant, and cluster uncertainty by old and new constituency. Conservative expenditure is lower by £2,645 per old constituency deprivation standard deviation (95% CI −£3,349 to −£1,942, $N = 1{,}028$ dyads), and Liberal Democrat expenditure by £1,552 (95% CI −£2,063 to −£1,041, $N = 1{,}036$ dyads). The Reform absolute expenditure gradient is imprecise at £106 (95% CI −£234 to £446, $N = 973$ dyads). Labour's total pound gradient is also imprecise, whereas its spending as a share of the legal limit has a positive association with deprivation. Party differences and measurement units therefore matter for interpreting relative allocation. These are expenditure levels across constituencies in 2024, rather than estimates of historical spending cuts.

Historical electoral opportunity helps explain this allocation geography. Appendix Table A29 compares base and adjusted expenditure per elector models on the common sample of 6,726 respondents, with 299 old origins and 380 new destinations. This exploratory extension was specified after the support and allocation patterns were known. The adjustment adds each party's notional 2019 vote share, its square and its shortfall from the notional winner, using old constituency fixed effects and two way clustered inference throughout. These historical measures describe competitive opportunity, with zero shortfall for the notional winner. They enter an allocation comparison rather than an adjustment to the total contextual support effect.

The comparisons in Appendix Table A29 show how historical electoral opportunity relates to the allocation gradient. For the Conservatives, adjustment changes the gradient from −£0.04268 to −£0.01979 per elector per old constituency deprivation standard deviation. The Liberal Democrat gradient changes from −£0.02061 to an imprecisely estimated £0.00282 in the same unit. Because these comparisons use identical respondents, their paired changes directly assess the shift associated with adding the opportunity measures. Both parties' opportunity blocks and paired slope changes survive the declared Holm correction across

four tests, whose full statistics and fixed urbanity sensitivity remain in the appendix table. Historical electoral position thus accounts for part of the observed allocation geography, while the remaining negative Conservative association is conditional on those measured proxies. Inferring strategists' motives or a causal deprivation penalty would require a design for allocation itself.

The connection between allocated effort and voters' encounters is less direct. Recalled contact and the selected digital targeting evidence retained in the appendix describe different stages of communication. The archived Google measures record intended geographic opportunity from selected central advertisers, rather than expenditure or delivered impressions. Conservative binary reach fails its declared multiple testing correction, and the duration bounds allow either direction for its association, while the Labour results also contain nondetections. Contact questions refer to the preceding 4 weeks at W27, the preceding 7 days at W28 and the campaign retrospectively at W29. Their timing comparisons therefore combine overlapping recall windows with uncertain receipt, and donor based constituency contact measures add sampling noise. The lower Conservative and Liberal Democrat contact in the original W26 to W28 comparison accompanies discordant and imprecise results in the routing and timing checks. These findings leave the transition from intended targeting to delivered and remembered communication unresolved.

Campaign content gives a complementary description of what parties communicated in the constituencies covered by the archive. Appendix Tables A40 and A41 report the OpenElections human coding of 1,543 leaflets from five parties in 306 English constituencies (Milazzo and Trumm, 2024). Labour immigration references increase with deprivation in this material, while the Conservative estimate is imprecise. These cross constituency associations concern selected archived leaflets and preserve the distinction between content availability and receipt by panel respondents. Narratives of neglect could become more relevant or credible in constituencies of lower economic standing, but the content evidence measures a communication environment rather than individual narrative exposure.

Perceived electoral opportunity provides a positive diagnostic on the voter side of this environment. Appendix Table A38 reports respondents' perceived chances of a Conservative victory between W27 and W28. The estimate is −1.57 points on the 0 to 100 chance scale per old constituency deprivation standard deviation (95% CI −2.74 to −0.39, $N = 12{,}798$). This wider eligible sample differs from the primary support frame, and the measure records perceived chances rather than factual spending change. It connects reassignment to a judgement relevant to competition while leaving the causal sequence from resources through perceptions to vote choice open.

Electoral trajectories help locate the support change, with exploratory status apart from the previously specified Labour to Reform transition. Appendix Table A36 shows lower Conservative voting among baseline Reform intenders and undecided respondents assigned to more deprived constituencies. The pooled resource comparisons retain uncertainty about Conservative recapture, and direct retention versus entry comparisons leave retention dominance unresolved. Some final interval resource contrasts survive their existing correction, including comparisons with campaign departures, while contrasts with earlier departures and entry remain imprecise. Persistence, entry and weighted contributions thus describe distinct features of the observed trajectories. The reported vote analyses condition on recorded voters. Separate actual ballot sensitivities include observed nonvoters, retain positive point estimates and fail their declared two test correction. Their different risk sets support neither a paired attenuation comparison nor a population turnout conclusion.

The remaining diagnostics constrain specific interpretations of the context effect rather than select a default psychological explanation. Appendix Table A44 and Figure A21 retain the measured neglect, attention, personal insecurity and spending adjustment results. These partial constructs allow psychological interpretation and political supply to operate together, while adjustment for spending after assignment remains descriptive (Acharya et al., 2016). The immigration priority interactions differ between the W25 human coding and W26 coding, but wave and coding change together. The paired interaction difference fails its multiple

testing correction, leaving the discrepancy unresolved rather than identifying coding as its cause. The tactical voting and origin incumbency sensitivities remain in Appendix Tables A38 and A39. Taken together, the allocation evidence identifies an observed feature of the electoral environment accompanying the conditional contextual effect, while causal transmission through resources, communication and narrative remains open.

Conclusion

My results show that constituency economic context affects radical right support in an electoral system that organizes competition by district. The boundary review in England assigned respondents from the same old constituency to new constituencies of different income deprivation. Comparing their changes in Reform UK support identifies a contextual effect under conditional parallel change.

The substantive change is modest for a typical reassignment but meaningful within the campaign. A shift of 1 standard deviation in respondents' actual exposure corresponds to about 1.17 percentage points of Reform support, against mean support of 15.0% before the campaign and 16.6% after the election in the primary sample. The contextual pattern extends to respondents with secure personal economic circumstances, although subgroup uncertainty leaves differences across economic positions open. Its political significance concerns the electoral environment built around voters, with constituency economic standing giving potential relevance to appeals about neglect and relative disadvantage.

The allocation findings give this interpretation an observable political foundation. Conservative and Liberal Democrat resources are lower relative to Reform in more income deprived constituencies on a common respondent comparison. Historical electoral opportunity accounts for part of that geography, leaving a smaller negative Conservative gradient and an imprecise adjusted Liberal Democrat gradient. Exploratory trajectories provide evidence about persistence and entry, while discordant resource and contact findings leave transmission to votes unresolved. Contextual demand and political supply can therefore operate together in the observed environment, with narrative persuasion remaining a theoretical interpretation rather than an identified pathway.

These findings concern one boundary review, one UK general election and an online panel in England. Their wider relevance lies in systems where constituency economic standing shapes the competitive opportunities facing parties and voters. Comparisons across campaigns

and electoral institutions could establish how that organization changes the consequences of economic geography. Tracing delivered communication and subsequent responses would connect the observed allocation of political effort more closely to the contextual effect on radical right support.

Data Availability

The analysis uses the British Election Study Internet Panel combined waves 1 to 30, version 30.1 (The British Election Study Team, 2025). The panel is available through the British Election Study website (`https://www.britishelectionstudy.com/data/`) subject to its access conditions, and through the UK Data Service under study SN 8202 and the End User Licence. The underlying respondent data remain subject to the providers' redistribution restrictions. Constituency boundaries, geographic lookups and deprivation inputs come from the Office for National Statistics, the House of Commons Library and the English Indices of Deprivation 2019. Campaign expenditure comes from the Electoral Commission, candidate records from Democracy Club, and archived leaflet content from OpenElections. The geographic inputs contain ONS and OS data, © Crown copyright and database right 2024, under the Open Government Licence v3.0. Replication materials will be deposited on acceptance, subject to the source licences.

Conflict of Interest

The author declares no competing interests.

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Appendix

1. **Geographic construction and coverage.** Small area coverage, old and new constituency coverage, official overlap coverage, boundary fallback cases, and independent validation of constituency deprivation ranks against the House of Commons record.

2. **Sample flow and outcome coding.** Sample restrictions for every endpoint, stable assignment coding, and outcome definitions. No sample rule uses the 2024 winner.

3. **Complete Reform event study.** All period coefficients with pointwise and simultaneous intervals and the joint test for prior trends.

4. **Predetermined covariate balance.** All balance models with adjusted values, the omnibus test, and same sample adjusted comparisons.

5. **Local perceptions.** Item construction, response availability, balance, equivalence tests, long service trajectories, and later outcomes.

6. **Missing data.** Retained sample estimates with missingness indicators and each feasible same sample complete case contrast.

7. **Alternative exposures and samples.** Employment deprivation, overall deprivation, overlap restrictions, candidate presence, winsorization of the shift, and outcome placebos.

8. **Shape of the response.** Signed categories, categorical event study, and flexible spline estimates.

9. **Attrition and weights.** Observation modeling, weight support and winsorization, extreme allocations, median dose group trimming with continuous slope reestimation, and the available longitudinal weight sensitivity.

10. **Conditional binary estimators.** Overlap weighting and augmented estimates with common support, clipping, effective sample size, and balance.

11. **Inference and influence.** One way and two way clustering, bias reduced CR2 with effective clusters, wild cluster bootstrap, randomization inference, and single constituency deletion.

12. **HonestDiD sensitivity (Rambachan-Roth).** Relative-magnitude and smoothness breakdown values for short (W19-W25) and long (W1-W25) pre-periods, with RM 95% CIs at $\bar{M} = 0$ and $\bar{M} = 0.25$.

13. **Specification curve and multiverse.** Covariate subset curve and vibration plot across treatment measures and adjustment sets. The primary estimate is highlighted.

14. **Residential mobility.** Never-changed codes, strict stability, renter and tenure exclusions, local authority stability, and the combined restriction.

Panel A. Construction and coverage diagnostics.

Table A1: Geographic coverage and validation against Commons deprivation ranks

Diagnostic	Count
IMD 2019 English LSOAs	32,844
LSOAs mapped to old constituency	32,844
old assignments using nearest-boundary fallback	5
LSOAs mapped to new constituency	32,844
old English constituencies	533
new English constituencies	543
official English old/new overlap pairs	1,165
overlap pairs with complete income shift	1,165
LSOA 2011/2021 changed geography	1,106

Panel B. Validation Spearman correlations against the House of Commons deprivation record.

Constructed measure	Commons rank	Spearman ρ	p
Income deprivation rate	Income	-0.994	<0.001
Employment deprivation rate	Employment	-0.993	<0.001
IMD score	IMD rank 2019	-1.000	<0.001

Panel A. Sequential restrictions (primary Reform W26 to W29 sample).

Table A2: Sample flow and outcome eligibility

Restriction	N	Old	New	Dyads
Valid panel identifier	122,382	632	632	1218
Official English old AND new constituency (W26)	25,038	533	543	1031
Verified official overlap pair (W26)	25,036	533	543	1029
Valid W26 and W29 Reform outcome	13,102	533	543	978
Stable observed geography at both endpoints	13,028	533	543	976
Old origin with within-origin dose variation (full sample)	8,766	343	423	786
Stable observed geography, W19 to W29	7,685	334	418	764

Panel B. Outcome-specific eligibility (stable English official pair with both outcomes).

Outcome	Waves	Elig. N	Old	New	Dyads	More dep. N	Less/eq. N
Reform support	W26-W29	11,802	533	543	963	5,131	6,671
Reform like	W26-W29	13,232	533	543	977	5,731	7,501
Reform PTV	W26-W30	3,137	531	542	796	1,335	1,802
Political attention	W26-W29	16,266	533	543	1002	7,064	9,202
Turnout likelihood	W26-W28	13,689	533	543	978	5,937	7,752

Table A3: Event study estimates for Reform support, all waves

Wave	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p
W19	-0.90	1.46	[-3.77, 1.97]	0.539
W20	-1.34	1.50	[-4.28, 1.60]	0.373
W21	-0.08	1.38	[-2.79, 2.62]	0.953
W22	-0.35	1.16	[-2.62, 1.93]	0.765
W23	-0.21	1.24	[-2.63, 2.22]	0.868
W25	-0.87	1.03	[-2.89, 1.16]	0.402
W27	1.31	0.93	[-0.51, 3.14]	0.157
W28	2.81	0.80	[1.24, 4.38]	<0.001
W29	2.51	0.87	[0.81, 4.20]	0.004
W30	1.91	1.22	[-0.48, 4.30]	0.118

Table A4: Balance of predetermined covariates on the deprivation shift

Covariate	N	Mean	SD	$\hat{\gamma}$	SE	p	Holm p
Age (W25)	5,070	59.872	13.921	0.028	0.487	0.954	1.000
Female	7,685	0.531	0.499	-0.017	0.016	0.273	1.000
Degree	4,977	0.337	0.473	-0.026	0.019	0.165	1.000
No qualifications	4,977	0.054	0.227	0.017	0.008	0.028	0.425
Household income band (W25, 1 to 15)	3,722	7.691	3.739	-0.555	0.159	<0.001	0.009
Personal income band (W25)	3,785	5.679	3.350	-0.354	0.134	0.009	0.137
Employed	5,070	0.454	0.498	-0.022	0.019	0.241	1.000
Unemployed	5,070	0.011	0.104	0.002	0.003	0.512	1.000
Homeowner	5,067	0.789	0.408	-0.028	0.014	0.049	0.631
Social renter	5,067	0.089	0.285	0.030	0.010	0.003	0.047
Non-white	5,017	0.047	0.212	0.000	0.007	0.990	1.000
Left to right self placement (W25)	4,431	4.974	2.202	-0.060	0.095	0.525	1.000
Reform party ID (W25)	4,924	0.027	0.163	-0.002	0.005	0.757	1.000
Reform support (W25)	4,467	0.079	0.270	0.003	0.008	0.708	1.000
EU Leave vote	6,899	0.486	0.500	0.022	0.016	0.168	1.000
Past Conservative vote (2019)	7,043	0.449	0.497	-0.007	0.015	0.650	1.000
Past Labour vote (2019)	7,043	0.324	0.468	-0.005	0.013	0.703	1.000
Past Reform/Brexit vote (2019)	7,043	0.021	0.144	0.008	0.004	0.040	0.565

Table A5: Estimates with adjustment for predetermined covariates

Spec.	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
unadjusted replication	2.51	0.87	[0.81, 4.20]	0.004	7,685
predetermined demographic and socioeconomic adjustment	2.42	0.87	[0.70, 4.13]	0.006	7,685
full predetermined adjustment including prior politics	2.35	0.85	[0.69, 4.01]	0.006	7,685
complete-case sample unadjusted	4.27	1.57	[1.19, 7.34]	0.007	2,435
complete-case sample fully adjusted	4.43	1.55	[1.39, 7.46]	0.004	2,435

Panel A. Balance of predetermined local-perception items on the deprivation shift D_i.

Table A6: Balance and tests for perceptions of the local area

Item	N	$\hat{\gamma}$	SE	p	Holm p
Local service cuts grievance (W25)	4,345	0.008	0.035	0.816	0.816
Perceived local disadvantage (W25)	752	0.202	0.050	<0.001	<0.001
Historical local service grievance	5,561	0.036	0.029	0.210	0.701
Area adverse, education (W25)	620	0.240	0.106	0.024	0.135
Area adverse, public spaces (W25)	781	0.171	0.075	0.022	0.135
Area adverse, shops (W25)	795	0.249	0.076	<0.001	0.008
Area adverse, crime (W25)	766	0.284	0.089	0.001	0.010
Area adverse, community spirit (W25)	779	0.119	0.089	0.182	0.701
Local economy deterioration (W23)	565	0.064	0.048	0.175	0.701

Panel B. Summary tests and perception-adjusted estimate.

Test	N	p	Conclusion
Local service cuts grievance (W25)	4,345	0.816	equivalent within 0.10 outcome standard deviations
Perceived local disadvantage (W25)	752	<0.001	equivalence remains inconclusive
Historical local service grievance	5,561	0.210	equivalence remains inconclusive
full predetermined and local perception adjustment, retained sample	7,685	0.005	effect remains after observed local perception adjustment
cuts not perceived as too far	707	0.022	descriptive scope estimate, not a mechanism test
more favorable half of area index	301	0.047	descriptive scope estimate, not a mechanism test
all local service perception leads jointly zero	6,806	0.307	trajectory equivalence remains inconclusive

Table A7: Alternative exposures, samples and outcome placebos

Outcome and specification	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
Reform support, income shift per 5pp	2.36	0.82	[0.76, 3.96]	0.004	7,685
Reform support, income shift per old-seat SD	2.51	0.87	[0.81, 4.20]	0.004	7,685
Reform support, old-overlap share at least 5%	2.48	0.89	[0.73, 4.22]	0.005	7,055
Reform support, old-overlap share at least 10%	2.74	0.99	[0.80, 4.67]	0.006	5,365
Reform support, destination had Reform candidate	2.66	0.89	[0.90, 4.41]	0.003	7,162
Reform support, absolute income shift at least 0.5pp	2.62	1.00	[0.66, 4.57]	0.009	3,299
Reform support, income-deprivation shift winsorized 1/99	2.73	1.05	[0.67, 4.79]	0.009	7,685
Reform support, W19 to W29 longitudinal weight sensitivity	1.21	1.48	[-1.70, 4.11]	0.416	3,924
Reform support, any more vs less or equally deprived destination	2.31	1.01	[0.32, 4.29]	0.023	4,938
Reform support, more vs less deprived destination, shift at least 0.5pp	2.51	1.46	[-0.36, 5.37]	0.087	2,400
Reform support, top vs bottom income-shift quartile	2.59	1.31	[0.02, 5.16]	0.048	2,969
Conservative support, party-outcome placebo family	-0.37	0.85	[-2.03, 1.29]	0.666	7,685
Labour support, party-outcome placebo family	-0.25	1.06	[-2.32, 1.83]	0.817	7,685
Lib Dem support, party-outcome placebo family	-0.27	0.73	[-1.70, 1.16]	0.708	7,685
Green support, party-outcome placebo family	-0.36	0.72	[-1.78, 1.06]	0.619	7,685
Reform support, party-outcome placebo family	2.51	0.87	[0.81, 4.20]	0.004	7,685
Non-vote, party-outcome placebo family	-0.63	0.48	[-1.58, 0.32]	0.196	7,685
Reform support (W26 level), baseline balance diagnostic	0.91	1.21	[-1.46, 3.28]	0.452	7,685
W29 Reform outcome (observed), differential attrition diagnostic	-1.69	1.09	[-3.82, 0.43]	0.119	14,250

Panel A. Signed categories relative to near-zero shifts.

Table A8: Shape of the response to the deprivation shift

Contrast	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
less deprived vs near-zero	-1.53	1.44	[-4.35, 1.28]	0.286	7,685
more deprived vs near-zero	1.44	1.32	[-1.15, 4.03]	0.276	7,685

Panel B. Spline-predicted change relative to zero boundary shift.

Shift (SD)	Shift (pp)	Pred. change (pp)	95% CI (pp)	p
-1.30	-6.89	9.62	[-2.58, 21.83]	0.122
-1.16	-6.14	4.25	[-2.85, 11.34]	0.241
-1.02	-5.39	0.63	[-4.21, 5.47]	0.799
-0.88	-4.64	-1.53	[-5.84, 2.79]	0.488
-0.73	-3.90	-2.51	[-6.51, 1.49]	0.219
-0.59	-3.15	-2.61	[-5.99, 0.77]	0.130
-0.45	-2.40	-2.13	[-4.72, 0.47]	0.108
-0.31	-1.65	-1.34	[-3.26, 0.58]	0.171
-0.17	-0.90	-0.55	[-1.88, 0.78]	0.416
-0.03	-0.15	-0.05	[-0.36, 0.27]	0.771
0.00	0.00	0.00	[0.00, 0.00]	—
0.11	0.59	0.13	[-1.00, 1.25]	0.826
0.25	1.34	0.36	[-1.53, 2.26]	0.708
0.39	2.09	0.67	[-1.55, 2.89]	0.552
0.53	2.84	1.03	[-1.46, 3.52]	0.419
0.68	3.59	1.40	[-1.56, 4.36]	0.355
0.82	4.33	1.76	[-1.82, 5.33]	0.335
0.96	5.08	2.07	[-2.03, 6.16]	0.322
1.10	5.83	2.30	[-2.04, 6.65]	0.299
1.24	6.58	2.43	[-1.95, 6.82]	0.277
1.38	7.33	2.43	[-2.39, 7.25]	0.323
1.52	8.05	2.27	[-4.44, 8.99]	0.507

Panel C. Less/more deprived vs near-zero by wave (post-treatment waves; reference W26).

Category	Wave	$\hat{\beta}$ (pp)	95% CI (pp)	p
more deprived	W27	0.55	[-2.18, 3.28]	0.692
more deprived	W28	1.73	[-0.81, 4.28]	0.182
more deprived	W29	1.44	[-1.15, 4.03]	0.276
more deprived	W30	0.59	[-4.09, 5.27]	0.804
less deprived	W27	-0.36	[-3.47, 2.75]	0.822
less deprived	W28	-0.93	[-3.72, 1.86]	0.513
less deprived	W29	-1.53	[-4.35, 1.28]	0.286
less deprived	W30	-2.38	[-6.79, 2.03]	0.289

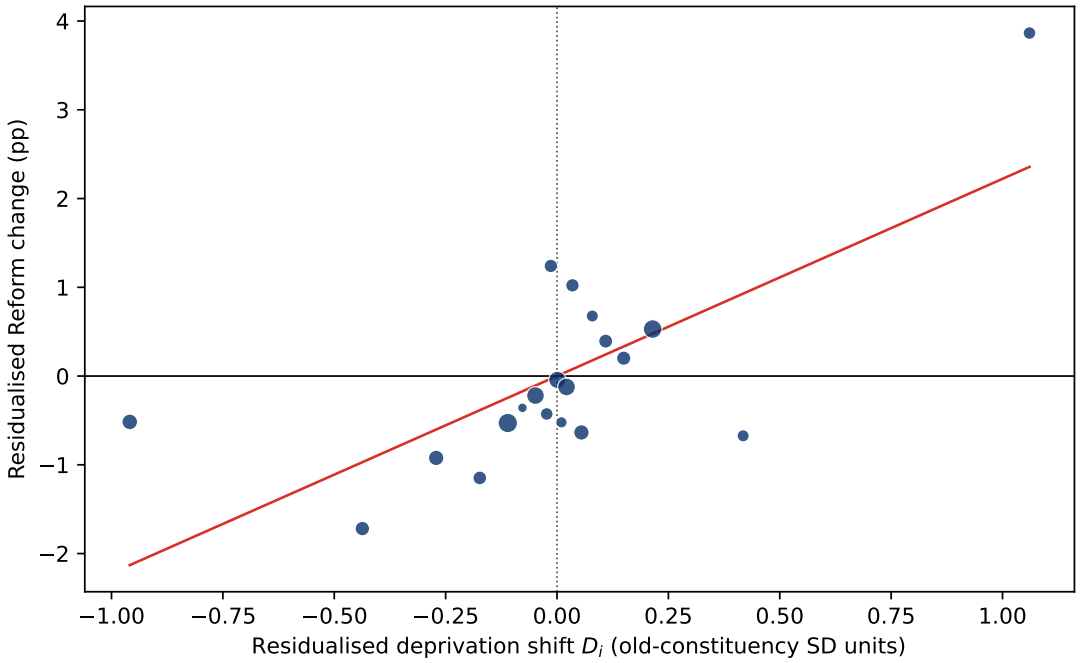


Figure A9: Change in Reform support by deprivation shift, binned means

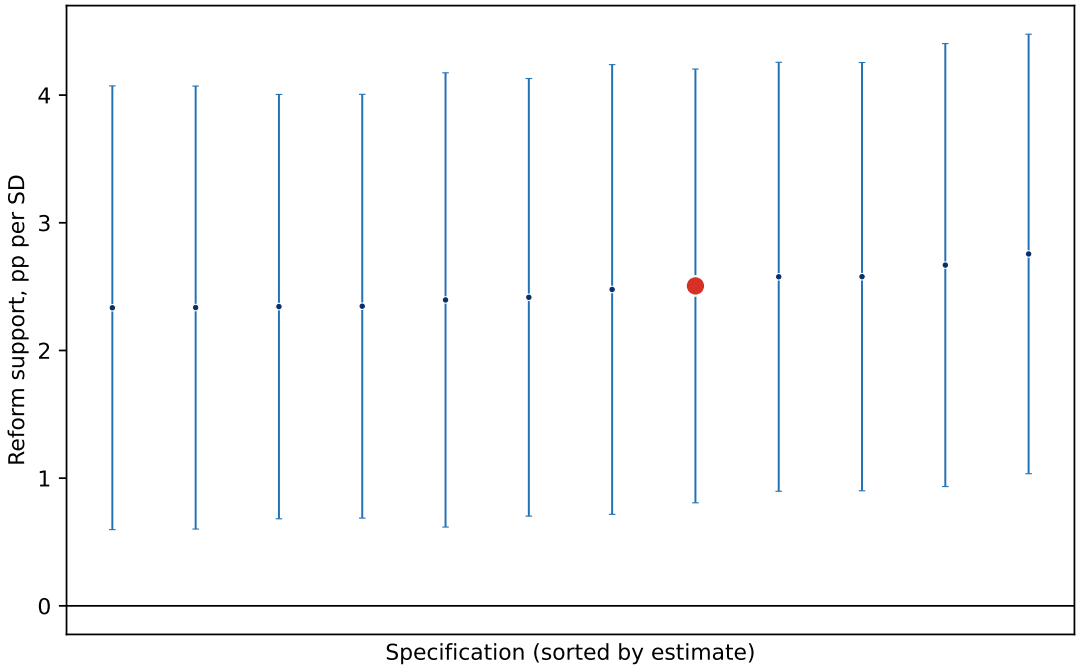


Figure A10: Estimates across 12 alternative specifications

Panel A. Clustering, CR2 and resampling.

Table A11: Alternative inference and influence of single constituencies

Spec.	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	Eff. clust.	N
Two-way old and new (primary)	2.51	0.87	[0.81, 4.20]	0.0038		7,685
One-way old origin	2.51	0.84	[0.85, 4.16]	0.0030		7,685
One-way new destination	2.51	0.69	[1.15, 3.86]	0.0003		7,685
CGM multiway (2-group, finite sample)	2.51	0.87	[0.81, 4.20]	0.0038		7,685
CR2 Bell-McCaffrey (old) + effective clusters	2.51	0.84	[0.85, 4.16]	0.0030	303.2	7,685
Wild cluster bootstrap Rademacher by old origin	2.51	—	—	0.0065		7,685
Randomization inference within old origin	2.51	—	—	0.0010		7,685

Panel B. Largest single-constituency deletion (fixed-projection).

Deletion	Constituency	$\Delta\hat{\beta}$ (pp)	Deleted N
origin	E14000643	-0.39	23
destination	E14001255	-0.29	15

Table A12: Sensitivity to violations of parallel trends

Target	Pre-period	RM breakdown $\bar{M}$	SM breakdown M (pp)	RM 95% CI at $\bar{M}$=0	RM 95% CI at $\bar{M}$=0.25
W28 only	short (W19 to W25)	0.49	0.225	[1.24, 4.38]	[0.61, 5.01]
W28 only	long (W1 to W25)	0.41	0.161	[1.25, 4.41]	[0.48, 5.17]
W29 only	short (W19 to W25)	0.21	0.073	[0.81, 4.20]	[-0.14, 5.15]
W29 only	long (W1 to W25)	0.17	0.079	[0.81, 4.20]	[-0.35, 5.36]
Mean W28 to W29	short (W19 to W25)	0.36	0.141	[1.12, 4.19]	[0.33, 4.98]
Mean W28 to W29	long (W1 to W25)	0.29	0.112	[1.13, 4.20]	[0.17, 5.17]
Mean W27 to W30	short (W19 to W25)	0.22	0.065	[0.69, 3.58]	[-0.09, 4.36]
Mean W27 to W30	long (W1 to W25)	0.18	0.061	[0.70, 3.53]	[-0.27, 4.49]

Table A13: Sensitivity to attrition

Analysis	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
unweighted W26 cohort complete cases	2.51	0.87	[0.81, 4.20]	0.004	7,685
inverse-probability-of-observation weighted	2.48	0.92	[0.68, 4.27]	0.007	7,685
IPOW winsorized 1/99	2.49	0.92	[0.69, 4.28]	0.007	7,685
extreme lower allocation	-34.34	1.25	[-36.78, -31.89]	<0.001	8,099
extreme upper allocation	36.34	1.23	[33.94, 38.75]	<0.001	8,099

Table A14: Estimates contrasting a more deprived with a less or equally deprived destination

Estimator	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
overlap-weighted old-origin-FE contrast	2.08	1.04	[0.04, 4.12]	0.046	4,938
augmented inverse-probability weighted conditional ATE	2.07	1.09	[-0.07, 4.20]	0.058	4,938
AIPW conditional ATE on propensity 0.05-0.95 common support	1.87	1.23	[-0.55, 4.29]	0.130	4,459

Panel A. Covariate-subset specification curve.

Table A15: Specification curve and multiverse summary

Spec.	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
Unadjusted	2.51	0.87	[0.81, 4.20]	0.004	7,685
Predetermined demographics	2.42	0.87	[0.70, 4.13]	0.006	7,685
Predetermined politics	2.38	0.84	[0.74, 4.01]	0.004	7,685
Predetermined full set	2.35	0.85	[0.69, 4.01]	0.006	7,685
Economic perceptions	2.57	0.86	[0.88, 4.26]	0.003	7,685
Service-cuts grievance	2.51	0.87	[0.82, 4.21]	0.004	7,685
Area and history	2.51	0.87	[0.80, 4.22]	0.004	7,685
Full retained set	2.39	0.85	[0.72, 4.07]	0.005	7,685

Panel B. Multiverse vibration summary.

Summary	Value
Specifications in grid	12
Range of estimates	[2.33, 2.76] pp
Significant at 5% (two-way)	12 of 12
Primary specification (income, unadjusted)	2.51 pp

Table A16: Continuous slope estimates after median dose group trimming.

Sensitivity	$\hat{\beta}$ (pp)	SE (pp)	95% CI (pp)	p	N
Primary (untrimmed)	2.51	0.87	[0.81, 4.20]	0.004	7,685
Highest outcome changes trimmed	2.48	0.86	[0.79, 4.17]	0.004	7,682
Lowest outcome changes trimmed	2.50	0.87	[0.80, 4.20]	0.004	7,681

Table A17: Distribution of identifying variation in the deprivation shift

Measure	Value
Share of within-origin variance from low-variance origins	0.037
Share of within-origin variance from high-variance origins	0.963
Low-variance origins (N)	167
High-variance origins (N)	167
Mean change near zero ($\lvert z \rvert \leq 0.25$)	1.03 pp
Mean change away positive ($z \geq 0.5$)	3.89 pp
Mean change away negative ($z \leq -0.5$)	3.11 pp

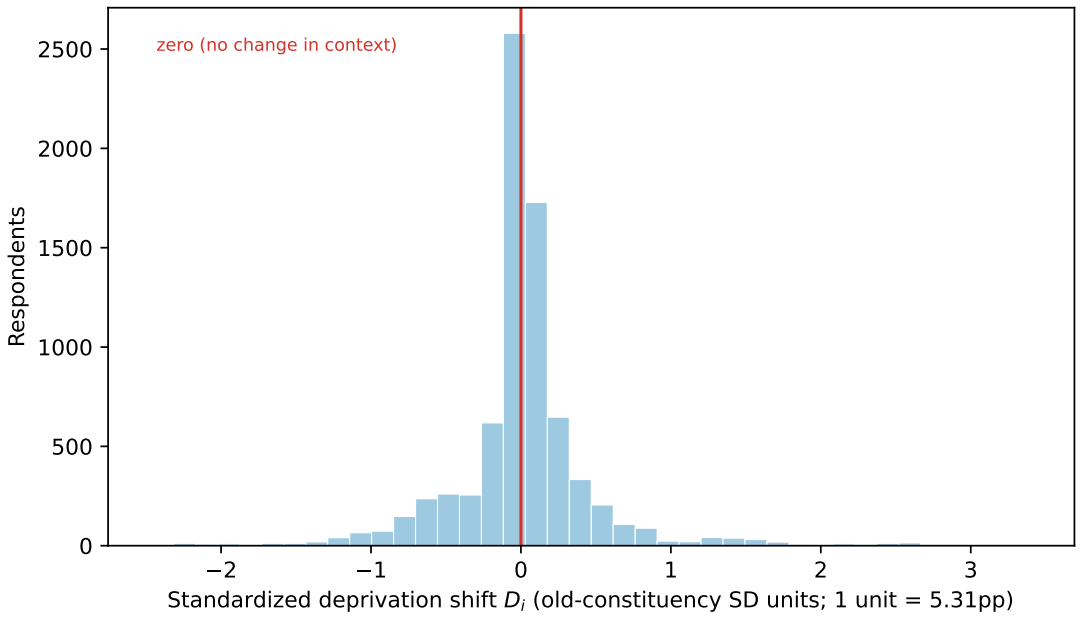


Figure A18: Distribution of the deprivation shift in the primary sample

Table A19: Summary statistics for the primary sample

Variable	Mean	SD	N
Reform support (W26)	0.150	0.357	7,685
Reform support (W28)	0.180	0.385	6,610
Reform support (W29)	0.166	0.372	7,685
Political attention (W26, 0 to 10)	7.007	2.172	7,664
Political attention (W29, 0 to 10)	6.794	2.195	7,659
Turnout likelihood (W26, 1 to 5)	4.763	0.743	7,649
Turnout likelihood (W28, 1 to 5)	4.850	0.553	5,485
Deprivation shift D_i (old-seat SD units)	0.010	0.466	7,685
Deprivation shift (pp income deprivation)	0.052	2.471	7,685
Age (W25)	59.872	13.921	5,393
Female	0.531	0.499	7,685
Degree	0.337	0.473	5,310
Household income band (W25, 1 to 15)	7.691	3.739	4,172
Employed	0.454	0.498	5,393
Homeowner	0.789	0.408	5,389
Non-white	0.047	0.212	5,339
Reform support (W25)	0.079	0.270	4,830
EU Leave vote	0.486	0.500	7,001

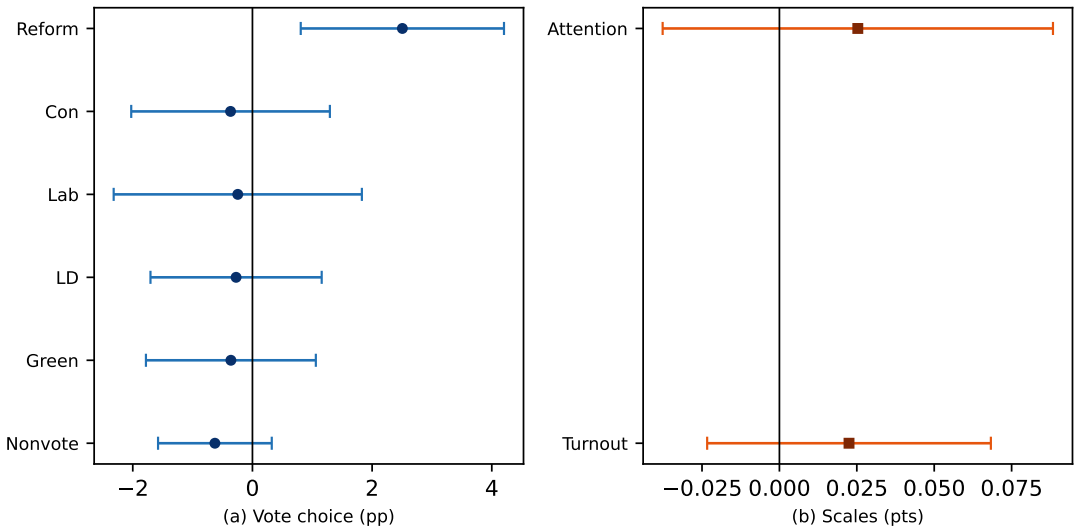


Figure A20: Effect of the deprivation shift on party support, nonvoting, attention and turnout

Table A20: Effect on Reform support by pre-campaign vote intention

Vote intention, W26	Estimate	SE	95% CI	p	N
Conservative	3.65	3.01	[-2.25, 9.54]	0.225	1,244
Labour	0.24	0.57	[-0.88, 1.36]	0.674	2,538
Other party	4.82	2.76	[-0.59, 10.23]	0.081	882
Would not vote	-11.67	11.75	[-34.71, 11.37]	0.321	20

Table A21: Effect of the deprivation shift on party support, nonvoting, attention and turnout

Outcome	$\hat{\beta}$	95% CI	p	N
Reform	2.51	[0.81, 4.20]	0.004	7,685
Con	-0.37	[-2.03, 1.29]	0.666	7,685
Lab	-0.25	[-2.32, 1.83]	0.817	7,685
LD	-0.27	[-1.70, 1.16]	0.708	7,685
Green	-0.36	[-1.78, 1.06]	0.619	7,685
Nonvote	-0.63	[-1.58, 0.32]	0.196	7,685
Attention	0.025	[-0.038, 0.088]	0.431	11,034
Turnout	0.022	[-0.023, 0.068]	0.336	9,089

Supplementary Validation

The supplementary evidence retains distinct sample histories alongside the current primary comparison. The original endpoint stable W26 and W29 sample contained 8,766 respondents and gave 2.18 percentage points of additional Reform support per old constituency deprivation standard deviation. The current primary sample contains 7,685 respondents whose old and new constituency codes remain unchanged across their observed waves from W19 to W29, and gives 2.51 points. The restriction rule preceded inspection of its own estimate, but its promotion to primary followed observation of both estimates. The comparison therefore records an outcome informed sample decision, as well as a measurement motivated restriction. Exposure units throughout are 1 cross sectional old constituency standard deviation, equivalent to 5.31 percentage points of the income deprivation rate, rather than 1 standard deviation of respondents' shifts. Each supplementary diagnostic retains its own eligible sample and outcome window.

Because the first-difference design removes stable individual levels, a baseline difference in income, local conditions, or political outlook matters for identification only when it predicts a different untreated change in Reform support after W26. Prior outcome dynamics, prior local service dynamics, and the stability of adjusted estimates directly evaluate this possibility.

The local perception analysis applies the same logic to prior service conditions. W25 perceived local service cuts are equivalent within a 0.10 outcome standard deviation margin, with an assignment coefficient of 0.0081 on the 1 to 5 scale, two-way clustered SE 0.0346, 95% CI -0.0598 to 0.0759, $p = 0.816$, TOST $p = 0.012$, $N = 4{,}345$. The historical local service grievance index narrowly fails the same equivalence test at the 5% level. The estimate is 0.036 index points, SE 0.029, 95% CI -0.020 to 0.093, $p = 0.210$, TOST $p = 0.058$, $N = 5{,}561$.

Service perceptions also followed comparable prior trajectories. The 11 lead coefficients are jointly small, Wald $\chi^2 = 12.79$, $df = 11$, $p = 0.307$, $N = 6{,}806$. The maximum simultaneous 90% bound exceeds the equivalence margin, leaving the stricter trajectory

equivalence assessment unresolved. Although the stricter equivalence assessment remains unresolved, the conventional joint evidence reveals no differential service grievance trajectory before the election.

The smaller W25 area quality battery identifies a baseline level difference. Respondents assigned to more deprived destinations report a higher area disadvantage index before the campaign. This difference would threaten the first-difference design only if it generated a differential untreated political trend, and the Reform event study and local service trajectory show comparable prior changes. Adjustment for predetermined characteristics, economic perceptions, W25 service cuts, the area index, the historical index, and missingness indicators yields 2.39 percentage points, two-way clustered SE 0.85, 95% CI 0.72 to 4.07, $p = 0.005$, $N = 7{,}685$.

Because the survey modules have an empty full complete case intersection, the comprehensive adjustment uses median filling and explicit missingness indicators. The feasible same-sample unadjusted and adjusted comparisons make the role of module coverage transparent. Across the observed adjustments that can be estimated, the Reform coefficient remains stable.

The outcome comparisons locate Reform support within the recorded voter sample. The Conservative, Labour, Liberal Democrat, Green and nonvoting estimates from that conditional frame remain imprecise, and their interpretation follows the frame's restriction to recorded post election voters. Political attention changes by 0.025 scale points, SE 0.032, 95% CI -0.038 to 0.088, $p = 0.431$, $N = 11{,}034$. Turnout likelihood changes by 0.022 points, SE 0.023, 95% CI -0.023 to 0.068, $p = 0.336$, $N = 9{,}089$. These confidence intervals quantify uncertainty, and the broader pattern makes generic political engagement a less compelling account of the Reform effect.

The geographic exposure covers all 32,844 English 2011 LSOAs and all 1,165 official English old and new overlap pairs; five coastal or generalized boundary centroid assignments

require a documented nearest polygon fallback. The reconstructed old constituency ranks are independently validated against the House of Commons deprivation record, while the main result is similar when employment deprivation or overall deprivation replaces the income deprivation measure and under overlap restrictions, a destination Reform candidate restriction, and winsorization of the deprivation shift. The primary sample uses stable codes across observed waves, while a separate sensitivity requires matching valid old and new codes at both W19 and W29. With valid matching W19 and W29 codes, the estimate is 2.60 points, SE 1.15, 95% CI 0.35 to 4.85, $p = 0.024$, $N = 3{,}954$ (Appendix Table A22). Respondents whose constituency codes never change from W19 to W29 give 2.51 points, SE 0.87, 95% CI 0.81 to 4.20, $p = 0.004$, $N = 7{,}685$, and dropping W26 renters gives 2.68 points, SE 0.78, 95% CI 1.16 to 4.20, $p < 0.001$, $N = 7{,}133$.

Table A22: Observed geographic stability

Row	$\hat{\beta}$	SE	95% CI	p	N	Old seats	Proxy coverage
R0 Original endpoint stability sample	0.022	0.008	[0.006, 0.037]	0.006	8,766	343	1.000
R1 Old and new codes unchanged at observed waves, W19 to W29	0.025	0.009	[0.008, 0.042]	0.004	7,685	334	1.000
R2 Valid old and new codes agree at W19 and W29	0.026	0.011	[0.003, 0.048]	0.024	3,954	297	1.000
R3 Excluding W26 renters	0.027	0.008	[0.012, 0.042]	<0.001	7,133	336	0.997
R4 Local authority unchanged at observed waves, W19 to W29	0.023	0.008	[0.007, 0.039]	0.005	7,943	338	1.000
R5 Tenure unchanged at observed waves, W19 to W29	0.023	0.009	[0.006, 0.041]	0.009	7,280	337	0.999
R7 R1, R3 and R5 combined	0.028	0.009	[0.009, 0.046]	0.003	5,396	321	0.997

Appendix Table A17 reports the treatment-variance decomposition weight diagnostic,

and Figure A9 shows the 20-bin residualized scatter. Together, these exhibits assess whether the continuous primary estimand is an informative summary. Relative to shifts within plus or minus 0.5 deprivation percentage points, the more deprived category estimate is 1.44 points, SE 1.32, 95% CI −1.15 to 4.03, $p = 0.276$, and the less deprived category estimate is −1.53 points, SE 1.44, 95% CI −4.35 to 1.28, $p = 0.286$. Their wide intervals leave a sharp threshold unresolved. The signed categories and flexible spline remain compatible with a graded relationship summarized by the continuous within-origin estimate. The linear slope weight and the scatter's visual linearity support the linear summary of the signed response.

The estimate is stable across constituency influence checks. Deleting one old origin changes the primary coefficient by at most 0.39 percentage points, and deleting one destination changes it by at most 0.29 points. A restricted Rademacher wild cluster bootstrap by old origin gives $p = 0.0065$ with 3,999 repetitions. Bell-McCaffrey (Bell and McCaffrey, 2002) CR2 by old origin and Carter-Schnepel-Steigerwald (Carter et al., 2017) effective clusters are reported in Appendix Table A11. With 334 nominal origins the correction is small and the CR2 probability value remains comparable to the two-way $p = 0.004$, reinforcing the primary two-way clustered inference. The overlap weighted comparison contrasts more deprived destinations with less or equally deprived destinations within old origins, changing the continuous treatment to a binary contrast.

Among the W26 cohort in identifying origins, 65.2% have a geographically consistent W29 Reform outcome; the W29 observation diagnostic is −1.69 percentage points, two-way clustered SE 1.09, 95% CI −3.82 to 0.43, $p = 0.119$. Inverse probability of observation weighting gives 2.48 points, SE 0.92, 95% CI 0.68 to 4.27, $p = 0.007$, $N = 7{,}685$. Winsorizing the weights at the 1st and 99th percentiles gives a closely similar estimate, while extreme missing outcome allocations span approximately −34.3 to +36.3 points (Manski, 1990). They are intentionally uninformative, demonstrating that tight attrition bounds require assumptions about selection. Appendix Table A16 reports a median dose group trimming sensitivity for the continuous slope. Respondents are divided at the cohort median dose,

and the excess observed share in the group above the median determines how much of that group's outcome change distribution is trimmed. The continuous slope is reestimated after removing either the highest or the lowest changes in Reform support from that group. The reported intervals describe ordinary two way clustered uncertainty for each refitted slope, rather than confidence intervals for bound endpoints. Inspired by the selection logic of Lee (2009), this exercise assesses sensitivity to tail removal rather than estimating Lee's formal bounds on the binary treatment effect among respondents who would always be observed.

The available longitudinal BES weight targets W19 to W29 and reduces the complete case sample to 3,924 respondents; applying it as a sensitivity to the W26 to W29 comparison gives 1.21 points, SE 1.48, 95% CI −1.70 to 4.11, $p = 0.416$. This estimate provides a distinct population sensitivity, and its interval includes both zero and the primary estimate. The primary estimand is defined for panel members observed at W26 and W29 with stable geographic assignment, who are not a probability sample of the English electorate, and the unweighted estimate should be read as the effect in that population.

The conditional weighting estimates preserve the direction and approximate magnitude of the effect. For the binary more deprived destination sensitivity, overlap weighting with predetermined covariate weights gives 2.08 percentage points, two-way clustered SE 1.04, 95% CI 0.04 to 4.12, $p = 0.046$, $N = 4{,}938$, with an effective sample size of 2,844 (Appendix Table A14). The corresponding augmented estimate is 2.07 points, score SE 1.09, 95% CI −0.07 to 4.20, $p = 0.058$, $N = 4{,}938$. Raw propensity scores range from 0.013 to 0.996 and 1.15% require clipping at 0.03 or 0.97. These models provide a coarser binary sensitivity because they change both the treatment definition and the identifying comparison. These overlap weights and augmented inverse probability estimates preserve the within-old-origin comparison under limited overlap. A multiverse of 12 specifications (three treatment measures by four adjustment sets) is reported as a vibration plot in Appendix Figure A10. The highlighted primary specification (income, unadjusted, 2.51 points) locates the main estimate within this fixed grid of treatment measures and adjustment sets.

Informativeness of the pre-period. Small leads are reassuring only to the extent that the pre-period could have detected a violation large enough to matter. The maximum simultaneous 95% upper bound on an absolute pre-period coefficient is 5.40 percentage points, larger than the 2.51-point W29 estimate. Honest parallel-change bounds (Rambachan and Roth, 2023) measure how large a departure from parallel change after W26, relative to the changes observed before W26, would overturn the result (Table A12). Under the relative magnitudes restriction, the confidence interval for the pooled W28 and W29 target first includes zero at the breakdown value $\bar{M} = 0.36$, about 0.4 points per wave; the W20 coefficient of -1.34 points (two-way clustered SE 1.50, 95% CI -4.28 to 1.60, $p = 0.373$, $N = 7{,}685$) drives the W20-to-W21 step of about 1.3 points that sets this benchmark. In a reanalysis of 42 panel studies from three leading political science journals, the mean breakdown value is 0.33, the median among significant results is 0.10, and eight studies reject a null effect at $\bar{M} = 0.5$ (Chiu et al., 2026). The pre-period therefore detects only large violations, and the placebo elections and the sensitivity analysis of a 2024 swing in the main text carry the weight of the argument.

The maximum across leads equivalence diagnostic for Reform support fails, since the smallest common bound supported for every pre period coefficient is 3.80 percentage points, which is 1.43 times the pooled W28 and W29 estimate of 2.66 points. The favourable joint lead test and this failed equivalence diagnostic together mean that the pre-period neither proves nor refutes parallel change; it can rule out only violations larger than this bound.

Further Checks

Aggregate overlap measures distinguish the part of an old constituency assigned to a new seat from the additional territory forming that seat. Table A24 separates these two frozen deprivation aggregates on the same eligible respondent sample, with both measures expressed relative to the old constituency's income deprivation and divided by the same standard

Table A23: Sensitivity to an unobserved confounder, benchmarked against observed covariates (Cinelli and Hazlett 2020)

Bounds use conventional standard errors. Primary estimates elsewhere use clustering by old origin and new destination.

Adjusted Reform support, percentage points per old constituency deprivation SD. Conventional 95% confidence intervals.

Sample	Benchmark	Benchmark strength multiple k	Estimate (pp)	95% CI
Main (N 7,685)	Left to right self placement	1	2.30	[0.68, 3.92]
Main (N 7,685)	Left to right self placement	2	2.25	[0.63, 3.88]
Main (N 7,685)	Left to right self placement	3	2.21	[0.58, 3.83]
Don't know kept (N 9,207)	Left to right self placement	1	2.35	[0.73, 3.97]
Don't know kept (N 9,207)	Left to right self placement	2	2.28	[0.66, 3.90]
Don't know kept (N 9,207)	Left to right self placement	3	2.20	[0.58, 3.82]
Stricter bound using all 18 covariates combined				
Main (N 7,685)	All 18 covariates combined	1	1.44	[−0.16, 3.05]
Main (N 7,685)	All 18 covariates combined	2	0.53	[−1.06, 2.12]
Main (N 7,685)	All 18 covariates combined	3	−0.38	[−1.96, 1.19]
Don't know kept (N 9,207)	All 18 covariates combined	1	1.20	[−0.40, 2.80]
Don't know kept (N 9,207)	All 18 covariates combined	2	−0.04	[−1.61, 1.54]
Don't know kept (N 9,207)	All 18 covariates combined	3	−1.28	[−2.83, 0.27]
Robustness values after adjustment for all 18 covariates				
Main sample, RV($q = 1$) 3.3%, RV($\alpha = .05$) 1.0%; Don't know kept, RV($q = 1$) 3.1%, RV($\alpha = .05$) 1.0%.				

deviation of income deprivation across old constituencies. The overlap aggregate is shared by respondents with the same old and new codes, rather than measuring their own neighbourhood deprivation. The effect is 1.93 points for the overlap area (SE 0.90) and 0.51 points for the joined territory (SE 0.91, 95% CI −1.28 to 2.31), with $N = 5{,}978$ respondents. In the diagnostic reference sample of 8,102 respondents before the variation restriction, raw income deprivation levels in the overlap and joined territory correlate at 0.62. The smallest detectable effect is about 2.6 points per standard deviation of the joined territory deprivation shift in that reference sample, so the split separates the two components only imprecisely.

Table A25 reports how the Reform estimate varies with two measures of campaign attention, each row giving the change in the effect per standard deviation of the measure. The effect varies little with the change in Reform's spending share (−0.05 points per standard

Table A24: Constituency deprivation within joined territories

Reform support change, W26 to W29, percentage points per old constituency deprivation SD. Old origin fixed effects. Standard errors and 95% confidence intervals use clustering by old origin and new destination.

Measure	Estimate (pp)	SE	95% CI	p	N
Old and new constituency overlap	1.93	0.90	[0.17, 3.69]	0.031	5,978
Joined territory	0.51	0.91	[-1.28, 2.31]	0.575	5,978

For the detectable effect, the exposure SD is calculated across respondents in the joined territory sample.

Diagnostic	Value
Correlation of raw deprivation levels in the overlap and joined territory (dimensionless)	0.62
Smallest detectable effect (pp per sample SD of the joined territory deprivation shift)	2.64

deviation, SE 0.53, 95% CI −1.10 to 1.00, $N = 7{,}636$), and the point estimate of the effect rises with the share of respondents reporting contact from Reform or the Brexit Party (1.04 points per standard deviation, SE 0.55; 0.76, SE 0.54, with marginality controls). Because contact is self-reported, the contrast is descriptive, and the contact estimates are imprecise and their intervals include zero.

Table A25: Reform support interactions per standard deviation

Campaign measure	Estimate (pp)	SE	95% CI	p	N
Spending-share change	-0.05	0.53	[-1.10, 1.00]	0.924	7,636
Brexit contact	1.04	0.55	[-0.03, 2.11]	0.057	7,685
Brexit contact, with marginality controls	0.76	0.54	[-0.30, 1.83]	0.161	7,685

Table A26 reports a pre-specified heterogeneity check of whether the effect differs among the 599 respondents who liked Reform (6 or more on a 0 to 10 scale) at W26 but did not intend to vote Reform; the second row is the difference from all other respondents. The difference is −0.55 points (SE 5.36), against an effect of 1.93 points among everyone else. With a smallest detectable difference of about 15 points, the test is too imprecise to tell whether the effect differs among these respondents.

Table A26: Reform support by prior sympathy

Group or measure	Estimate (pp)	SE	95% CI	p	N
Everyone else	1.93	0.78	[0.40, 3.46]	0.013	6,427
Difference for prior Reform sympathisers	-0.55	5.36	[-11.06, 9.96]	0.919	6,427
80% MDE for the difference (pp)	15.02				
Prior Reform sympathisers					599

Analysis Plans

The main effect design, a first difference between W26 and W29 with old constituency fixed effects and two way clustered standard errors, predates the plans described here and was not preregistered. Two analysis plans for the mechanism tests were committed to a version controlled repository before any estimate they specify was computed. The first plan, committed on 9 October 2026, specifies the test of whether the effect is concentrated among respondents who already liked Reform. The second plan, committed on 12 October 2026, specifies the tests of perceived neglect, attention and campaign spending, and was extended by addenda committed between 12 and 16 October 2026 for the transition, recapture, issue ownership and latent voter tests. Each addendum was committed before the estimates it specifies.

Table A27 lists every mechanism test in the two plans, the date on which the plan section that specifies it was committed, and its status. The status follows the plans themselves, so that a test is prespecified where the plan labels it confirmatory or specifies it in advance as a discriminator, and exploratory where the plan labels it so. The plans label as exploratory every test whose minimum detectable effect, computed from simulated outcomes, exceeds 2.5 percentage points. The strategic voting diagnostic appears in neither plan and is exploratory. The broader political organization account developed after the contextual finding and the investigation of allocation. The later historical opportunity extension was specified after the support and allocation results were known, so its four test family and fixed urbanity sensitivity remain exploratory. These later analyses leave the dates and classifications of the

earlier plans intact.

Table A27: Mechanism tests in the analysis plans, by plan date and status

Test	Plan date	Status
Effect among W26 non intenders who liked Reform	9 Oct 2026	prespecified
Perceived neglect, politicians do not care, W26 to W27	12 Oct 2026	prespecified
Perceived neglect, trust in MPs, W26 to W29	12 Oct 2026	prespecified
Attention to rhetoric, interaction with dose	12 Oct 2026	prespecified
Interaction with dose by Con and LD competitiveness	12 Oct 2026	exploratory
Interaction with dose by incumbent on the ballot	12 Oct 2026	exploratory
Interaction with dose by W26 Con or LD vote intention	12 Oct 2026	exploratory
Immigration priority, interaction with dose	12 Oct 2026	exploratory
Reported party contact, own fragment against joined territory	12 Oct 2026	exploratory
Con+LD spending experienced (per SD)	12 Oct 2026	exploratory
Reform support with Don't know kept, W26 to W29	12 Oct 2026	prespecified
Inflow to Reform from Labour intenders	12 Oct 2026	prespecified
Inflow to Reform from other origins, retention and timing	12 Oct 2026	exploratory
Conservative recapture of W26 Reform intenders	12 Oct 2026	exploratory
Issue ownership and party leaflet content	16 Oct 2026	exploratory
Latent Reform support among available non intenders (pooled)	16 Oct 2026	exploratory
Strategic voting, mechanism figure	none	exploratory

Table A28: Economic position and political organization diagnostics

Panel A. Own economic position (pp per old-constituency SD unless noted).

Estimate	Est.	SE / 95% CI	p	N
Household income interaction (per band)	0.79	[-0.74, 2.32]	0.310	4,172
Joint test, 6 interactions ($\chi^2(6)$)	—	—	0.763	3,986
Estimate at P25 household income	2.63	[-0.33, 5.60]	0.082	4,172
Estimate at median household income	3.06	[0.30, 5.81]	0.030	4,172
Estimate at P75 household income	3.90	[0.87, 6.94]	0.012	4,172
Equivalence (TOST, half primary effect)	—	—	0.278	4,172
Upper-half household income (exploratory split)	5.70	[1.80, 9.59]	0.004	1,708

Panel B. Personal economic insecurity (native 0 to 10 scale).

Estimate	Est.	SE / 95% CI	p	N
W25 to W27 change (0 to 10 scale)	-0.05	SE 0.09	0.549 (TOST 0.008)	3,804
W25 to W30 change (0 to 10 scale)	-0.18	SE 0.10	0.062 (Holm 0.123)	3,407

Panel C. Weekly slopes, W26 to W28 Reform change (pp).

Estimate	Est.	SE / 95% CI	p	N
Week 1 (from 2024-06-10)	1.96	[-0.71, 4.62]	0.150	8,837
Week 2 (from 2024-06-17)	2.19	[-0.46, 4.85]	0.106	8,837
Week 3 (from 2024-06-24)	2.69	[0.50, 4.88]	0.016	8,837
Week 4 (from 2024-07-01)	2.70	[-1.22, 6.62]	0.177	8,837
Joint test, equal weekly slopes ($\chi^2(3)$)	—	—	0.975	8,837

Panel D. Campaign effort (contact in pp; spending in £ per SD).

Estimate	Est.	SE / 95% CI	p	N
Reported Reform contact, W26 to W28 (pp)	-0.31	[-1.95, 1.33]	0.709	6,568
Reform candidate spending (£, per SD)	106	[-234, 446]	0.542	973
Conservative candidate spending (£, per SD)	-2645	[-3349, -1942]	<0.001	1,028
Liberal Democrat spending (£, per SD)	-1552	[-2063, -1041]	<0.001	1,036
Reported contact, any party, W26 to W28 (pp)	-4.43	[-7.54, -1.32]	0.005 (Holm 0.074)	6,620

Panel E. Incumbency (pp).

Estimate	Est.	SE / 95% CI	p	N
Adjusted for incumbent loss (pp)	2.44	[0.78, 4.11]	0.004	7,685
Primary specification (for comparison, pp)	2.51	[0.81, 4.20]	0.004	7,685
Shift × incumbent on ballot interaction (pp)	0.87	[-3.72, 5.46]	0.711	7,685
Incumbent-loss association, exploratory (pp)	2.13	[0.35, 3.91]	0.019	7,685

Table A29: Historical allocation opportunity

Comparison	Estimate	SE	95% CI	p	Holm p	N
Exploratory allocation, GBP per elector. Two way old origin and new destination CR1 inference. Coefficient p values are descriptive.						
Conservative, base						
Dose, old constituency deprivation SD	-0.0427	0.00598	[-0.0544, -0.031]	9.51e-13		6,726
Conservative, opportunity adjusted						
Dose, old constituency deprivation SD	-0.0198	0.00638	[-0.0323, -0.00728]	0.00193		6,726
2019 Conservative share, fraction	0.961	0.396	[0.185, 1.74]	0.0152		6,726
Share fraction squared	-0.545	0.336	[-1.2, 0.113]	0.105		6,726
Conservative shortfall, share fraction	0.123	0.11	[-0.0921, 0.339]	0.262		6,726
Conservative, urbanity stability						
Dose, old constituency deprivation SD	-0.0199	0.00656	[-0.0328, -0.00708]	0.00236		6,726
2019 Conservative share, fraction	0.961	0.397	[0.184, 1.74]	0.0154		6,726
Share fraction squared	-0.542	0.337	[-1.2, 0.119]	0.108		6,726
Conservative shortfall, share fraction	0.125	0.111	[-0.0916, 0.342]	0.258		6,726
Urban minus rural indicator	0.00161	0.0082	[-0.0145, 0.0177]	0.845		6,726
Liberal Democrat, base						
Dose, old constituency deprivation SD	-0.0206	0.00369	[-0.0278, -0.0134]	2.3e-08		6,726
Liberal Democrat, opportunity adjusted						
Dose, old constituency deprivation SD	0.00282	0.00381	[-0.00465, 0.0103]	0.46		6,726
2019 Liberal Democrat share, fraction	1.32	0.158	[1.01, 1.63]	7.73e-17		6,726
Share fraction squared	-1.49	0.285	[-2.04, -0.927]	1.89e-07		6,726
Liberal Democrat shortfall, share fraction	0.0867	0.0389	[0.0105, 0.163]	0.0257		6,726
Liberal Democrat, urbanity stability						
Dose, old constituency deprivation SD	0.00378	0.0038	[-0.00366, 0.0112]	0.32		6,726
2019 Liberal Democrat share, fraction	1.29	0.159	[0.979, 1.6]	4.79e-16		6,726
Share fraction squared	-1.45	0.285	[-2.01, -0.894]	3.54e-07		6,726
Liberal Democrat shortfall, share fraction	0.072	0.0384	[-0.00329, 0.147]	0.0609		6,726
Urban minus rural indicator	-0.00788	0.00935	[-0.0262, 0.0104]	0.399		6,726
Paired adjusted minus base dose slopes, original four test Holm family						
Conservative	0.0229	0.00482	[0.0134, 0.0323]	2.03e-06	2.03e-06	6,726
Liberal Democrat	0.0234	0.0041	[0.0154, 0.0315]	1.14e-08	2.27e-08	6,726
All four primary tests, the declared four test Holm family						
Test	Wald χ^2	df		p	Holm p	N
Conservative opportunity block	56.9	3		2.68e-12	8.04e-12	6,726
Conservative paired change	22.6	1		2.03e-06	2.03e-06	6,726
Liberal Democrat opportunity block	123	3		1.38e-26	5.51e-26	6,726
Liberal Democrat paired change	32.6	1		1.14e-08	2.27e-08	6,726

Table A30: Contact timing and allocation diagnostics

Comparison	Estimate	SE	95% CI	p	Holm p	N
W27 recall covers four weeks, W28 seven days, W29 the campaign. Windows can overlap. Timing is not equal duration exposure. Two way old/new inference, old origin fixed effects.						
endpoint only, aligned N 7171, overlapping W28 and W27 windows N 292, gap days min/median/max 2.69/21/39.9						
Conservative, W27 contact, subsequent Reform change, family T	1.21	0.971	[-0.688, 3.12]	0.211	1	7,171
Conservative, W28 contact, preceding Reform change, family T	0.807	0.555	[-0.281, 1.9]	0.146	1	7,171
Conservative, Subsequent minus preceding, W27 contact, family P	1.6	1.41	[-1.16, 4.36]	0.255	0.766	7,171
Conservative, Contact per deprivation SD, family A	-2.46	0.835	[-4.1, -0.827]	0.00317	0.0127	7,171
Liberal Democrat, W27 contact, subsequent Reform change, family T	-1.36	0.792	[-2.91, 0.195]	0.0865	0.692	7,171
Liberal Democrat, W28 contact, preceding Reform change, family T	0.00731	0.577	[-1.12, 1.14]	0.99	1	7,171
Liberal Democrat, Subsequent minus preceding, W27 contact, family P	-1.95	1.15	[-4.2, 0.301]	0.0896	0.359	7,171
Liberal Democrat, Contact per deprivation SD, family A	-1.46	0.649	[-2.73, -0.185]	0.0248	0.0495	7,171
Labour, W27 contact, subsequent Reform change, family T	-0.926	0.843	[-2.58, 0.726]	0.272	1	7,171
Labour, W28 contact, preceding Reform change, family T	0.482	0.499	[-0.496, 1.46]	0.334	1	7,171
Labour, Subsequent minus preceding, W27 contact, family P	-0.937	1.26	[-3.41, 1.54]	0.458	0.917	7,171
Labour, Contact per deprivation SD, family A	-0.168	1.23	[-2.57, 2.24]	0.891	0.891	7,171
Reform, W27 contact, subsequent Reform change, family T	-4.9	3.59	[-11.9, 2.13]	0.172	1	7,171
Reform, W28 contact, preceding Reform change, family T	0.898	0.987	[-1.04, 2.83]	0.363	1	7,171
Reform, Subsequent minus preceding, W27 contact, family P	-3.34	5.34	[-13.8, 7.13]	0.532	0.917	7,171
Reform, Contact per deprivation SD, family A	0.796	0.306	[0.196, 1.4]	0.00934	0.028	7,171

Contact coefficients are percentage points per contact indicator, allocation slopes per deprivation SD. T temporal (8), P paired (4) and A allocation (4) families retain Holm p values within each geography, not a new pooled family. Recall and noisy donor shares do not establish receipt or mediation.

Table A31: Contact timing and allocation diagnostics continued

Comparison	Estimate	SE	95% CI	p	Holm p	N
W27 recall covers four weeks, W28 seven days, W29 the campaign. Windows can overlap. Timing is not equal duration exposure. Two way old/new inference, old origin fixed effects.						
observed to endpoint, aligned N 6324, overlapping W28 and W27 windows N 263, gap days min/median/max 2.69/21/39.9						
Conservative, W27 contact, subsequent Reform change, family T	1.64	1.03	[-0.391, 3.66]	0.114	0.91	6,324
Conservative, W28 contact, preceding Reform change, family T	0.872	0.619	[-0.341, 2.08]	0.159	0.994	6,324
Conservative, Subsequent minus preceding, W27 contact, family P	2.56	1.52	[-0.408, 5.54]	0.0909	0.363	6,324
Conservative, Contact per deprivation SD, family A	-2.69	0.926	[-4.51, -0.877]	0.00365	0.0146	6,324
Liberal Democrat, W27 contact, subsequent Reform change, family T	-1.08	0.986	[-3.01, 0.851]	0.273	1	6,324
Liberal Democrat, W28 contact, preceding Reform change, family T	-0.0681	0.636	[-1.32, 1.18]	0.915	1	6,324
Liberal Democrat, Subsequent minus preceding, W27 contact, family P	-1.46	1.33	[-4.06, 1.14]	0.271	0.813	6,324
Liberal Democrat, Contact per deprivation SD, family A	-1.39	0.658	[-2.68, -0.0964]	0.0352	0.0703	6,324
Labour, W27 contact, subsequent Reform change, family T	-1.21	0.827	[-2.83, 0.406]	0.142	0.994	6,324
Labour, W28 contact, preceding Reform change, family T	0.432	0.548	[-0.641, 1.51]	0.43	1	6,324
Labour, Subsequent minus preceding, W27 contact, family P	-1.15	1.26	[-3.61, 1.31]	0.36	0.813	6,324
Labour, Contact per deprivation SD, family A	-0.27	1.33	[-2.88, 2.34]	0.839	0.839	6,324
Reform, W27 contact, subsequent Reform change, family T	-5.71	4.1	[-13.7, 2.33]	0.164	0.994	6,324
Reform, W28 contact, preceding Reform change, family T	1.04	1.03	[-0.984, 3.07]	0.313	1	6,324
Reform, Subsequent minus preceding, W27 contact, family P	-2.68	5.94	[-14.3, 8.96]	0.652	0.813	6,324
Reform, Contact per deprivation SD, family A	0.788	0.327	[0.147, 1.43]	0.0159	0.0478	6,324
Contact coefficients are percentage points per contact indicator, allocation slopes per deprivation SD. T temporal (8), P paired (4) and A allocation (4) families retain Holm p values within each geography, not a new pooled family. Recall and noisy donor shares do not establish receipt or mediation.						

Table A32: Selected intended targeting and duration bounds

Comparison	Estimate	SE	95% CI	p	Holm p	N
historical endpoint. Selected central advertiser Google intended postal targeting, not expenditure, delivered impressions or receipt. Two way old/new inference. Original 80 member Holm family, blanks are outside it.						
Binary reach, Adjusted dose slope	2.3	0.845	[0.645, 3.96]	0.00647		8,766
Binary reach, Paired dose slope change	0.119	0.23	[-0.332, 0.57]	0.606	1	8,766
Binary reach, Allocation, Conservative	-3.08	1.21	[-5.45, -0.716]	0.0107	0.416	8,766
Binary reach, Allocation, Labour	-5.86	2.22	[-10.2, -1.51]	0.00825	0.338	8,766
Binary reach, Allocation, Liberal Democrat	-7.13	2.04	[-11.1, -3.14]	0.000462	0.0203	8,766
Minimum duration, Adjusted dose slope	2.33	0.846	[0.673, 3.99]	0.00585		8,766
Minimum duration, Paired dose slope change	0.147	0.225	[-0.294, 0.588]	0.513	1	8,766
Minimum duration, Allocation, Conservative	-0.239	0.0553	[-0.347, -0.131]	1.52e-05	0.00103	8,766
Minimum duration, Allocation, Labour	0.0334	0.0512	[-0.067, 0.134]	0.515	1	8,766
Minimum duration, Allocation, Liberal Democrat	-0.179	0.044	[-0.265, -0.0927]	4.73e-05	0.0026	8,766
Maximum duration, Adjusted dose slope	2.33	0.843	[0.676, 3.98]	0.00574		8,766
Maximum duration, Paired dose slope change	0.146	0.223	[-0.292, 0.584]	0.513	1	8,766
Maximum duration, Allocation, Conservative	-0.242	0.0552	[-0.35, -0.133]	1.2e-05	0.000909	8,766
Maximum duration, Allocation, Labour	0.0293	0.0542	[-0.077, 0.136]	0.589	1	8,766
Maximum duration, Allocation, Liberal Democrat	-0.177	0.0431	[-0.261, -0.0925]	4.06e-05	0.00252	8,766

Allocation units are reach percentage points for binary reach and within party seat SD per deprivation SD for duration. Adjusted and paired dose slopes are Reform percentage points per deprivation SD.

Duration outer slope envelopes, raw intended targeting days per deprivation SD, not confidence intervals or true delivery bounds

Party	Outer slope envelope
Conservative	[-72.8, 48]
Labour	[-675, 782]
Liberal Democrat	[-50, -6.27]

Table A33: Selected intended targeting and duration bounds continued

Comparison	Estimate	SE	95% CI	p	Holm p	N
longer observed. Selected central advertiser Google intended postal targeting, not expenditure, delivered impressions or receipt. Two way old/new inference. Original 80 member Holm family, blanks are outside it.						
Binary reach, Adjusted dose slope	2.66	0.922	[0.848, 4.46]	0.00399		7,685
Binary reach, Paired dose slope change	0.15	0.26	[-0.359, 0.659]	0.564	1	7,685
Binary reach, Allocation, Conservative	-3.28	1.25	[-5.74, -0.823]	0.00889	0.356	7,685
Binary reach, Allocation, Labour	-5.88	2.18	[-10.2, -1.6]	0.00711	0.298	7,685
Binary reach, Allocation, Liberal Democrat	-6.98	2.01	[-10.9, -3.04]	0.00051	0.0219	7,685
Minimum duration, Adjusted dose slope	2.54	0.933	[0.708, 4.37]	0.00657		7,685
Minimum duration, Paired dose slope change	0.0313	0.254	[-0.466, 0.529]	0.902	1	7,685
Minimum duration, Allocation, Conservative	-0.248	0.0567	[-0.359, -0.137]	1.25e-05	0.000922	7,685
Minimum duration, Allocation, Labour	0.0206	0.0489	[-0.0751, 0.116]	0.673	1	7,685
Minimum duration, Allocation, Liberal Democrat	-0.165	0.0412	[-0.246, -0.0846]	5.94e-05	0.00285	7,685
Maximum duration, Adjusted dose slope	2.54	0.931	[0.715, 4.36]	0.00637		7,685
Maximum duration, Paired dose slope change	0.034	0.252	[-0.46, 0.528]	0.893	1	7,685
Maximum duration, Allocation, Conservative	-0.25	0.0566	[-0.361, -0.139]	1e-05	0.000793	7,685
Maximum duration, Allocation, Labour	0.0166	0.0522	[-0.0857, 0.119]	0.75	1	7,685
Maximum duration, Allocation, Liberal Democrat	-0.164	0.0407	[-0.244, -0.0847]	5.28e-05	0.00279	7,685

Allocation units are reach percentage points for binary reach and within party seat SD per deprivation SD for duration. Adjusted and paired dose slopes are Reform percentage points per deprivation SD.

Duration outer slope envelopes, raw intended targeting days per deprivation SD, not confidence intervals or true delivery bounds

Party	Outer slope envelope
Conservative	[-72.6, 47]
Labour	[-677, 739]
Liberal Democrat	[-46.4, -5.81]

Table A34: Direct retention contrasts

Comparison	Estimate	SE	95% CI	p	Holm p	N
Direct paired contrasts, percentage points per GBP 0.10 per elector. Two way old/new clustered normal inference. Original source Holm families retained. Intervals have unequal durations, not causal retention effects.						
Final departure comparisons						
Endpoint geography, Final departure minus entry	1.3	1.17	[-0.99, 3.59]	0.266	0.879	3,711
Endpoint geography, Final departure minus precampaign departure	1.71	1.39	[-1.02, 4.44]	0.22	0.879	3,711
Endpoint geography, Final departure minus campaign departure	2.92	1.1	[0.765, 5.07]	0.00787	0.0394	3,711
Endpoint geography, Final Reform minus mainstream departure	5.41	1.93	[1.63, 9.19]	0.00498	0.0299	3,711
Longer observed geography, Final departure minus entry	1.77	1.24	[-0.66, 4.2]	0.153	0.606	3,331
Longer observed geography, Final departure minus precampaign departure	2.11	1.47	[-0.774, 5]	0.151	0.606	3,331
Longer observed geography, Final departure minus campaign departure	3.43	1.17	[1.14, 5.72]	0.0033	0.0198	3,331
Longer observed geography, Final Reform minus mainstream departure	5.97	2.11	[1.83, 10.1]	0.00473	0.0237	3,331
Mainstream intender retention minus Reform entry						
Endpoint geography, Mainstream retention minus Reform entry, W28	-1.55	2.83	[-7.1, 4]	0.585	1	1,694
Endpoint geography, Mainstream retention minus Reform entry, W29	0.222	2.75	[-5.17, 5.62]	0.936	1	1,694
Longer observed geography, Mainstream retention minus Reform entry, W28	0.354	2.9	[-5.32, 6.03]	0.903	1	1,494
Longer observed geography, Mainstream retention minus Reform entry, W29	0.332	2.9	[-5.36, 6.02]	0.909	1	1,494

Table A35: Actual ballot and paired issue priority sensitivities

Comparison	Estimate	SE	95% CI	p	Holm p	N
Actual ballot sensitivity. Known nonvoters are zero, unknown turnout and unknown vote remain unknown. Outcome percentage points per deprivation SD. Old origin FE and two way old/new inference. Original two test family B.						
endpoint only	1.55	0.784	[0.0143, 3.09]	0.0479	0.0868	9,854
observed through w29	1.75	0.867	[0.052, 3.45]	0.0434	0.0868	8,656
Identical 4,817 respondents with W25 and W26 priority measures in the primary W26/W29 Reform support frame and identifying old origins. Priority versus other issues, percentage point slope difference per old constituency deprivation SD. Two way old origin and new destination clustered inference. Original three test Holm family.						
W26 immigration priority, difference in deprivation slope	5.42	2.45	[0.628, 10.2]	0.0267	0.08	4,817
W25 immigration priority, difference in deprivation slope	0.517	3.33	[-6.01, 7.05]	0.877	0.877	4,817
W26 minus W25 immigration priority interaction	4.91	3.09	[-1.16, 11]	0.113	0.226	4,817

Table A36: Conservative recapture trajectories

Comparison	Estimate	SE	95% CI	p	Holm p	N
Conservative outcome among W26 Reform or undecided origins (P), not baseline mainstream intenders. Percentage points, two way old origin and new destination clustered inference. Resource coefficients use sample standardized spending, not GBP per elector. Dose uses old constituency deprivation SD. Exploratory, no Holm family assigned.						
Resource only, Conservative spending per sample SD	3.15	1.74	[-0.255, 6.55]	0.0698		2,559
Dose and resource, dose	-5.91	2.84	[-11.5, -0.343]	0.0375		2,559
Dose and resource, Conservative spending per sample SD	1.12	2.02	[-2.84, 5.08]	0.579		2,559
Dose only, dose	-6.56	2.51	[-11.5, -1.64]	0.00898		2,559

Table A37: Seat competitiveness and the deprivation shift

Estimates per old constituency deprivation SD. Standard errors and 95% confidence intervals use clustering by old origin and new destination.

Specification	Estimate	SE	95% CI	p	N
Panel A. Dose effect on the change in competitiveness (pp)					
Change in notional winner to runner up margin	-0.32	1.48	[-3.21, 2.58]	0.831	7,685
Conservative viable, change	2.68	1.18	[0.38, 4.99]	0.023	7,685
Runner-up changed	1.51	3.69	[-5.73, 8.75]	0.683	7,685
Panel B. Effect on Reform support (pp), competitiveness held constant					
Holding change in notional margin	2.46	0.81	[0.87, 4.05]	0.002	7,685
Holding change in Conservative viability	2.44	0.88	[0.71, 4.17]	0.006	7,685
Holding runner-up change	2.48	0.87	[0.78, 4.18]	0.004	7,685
Panel C. Candidate spending slope (£ per SD), constituency pieces with old-constituency fixed effects					
Conservative	-2,645	359	[-3,349, -1,942]	<0.001	1,028
Conservative + notional margin	-2,642	364	[-3,355, -1,929]	<0.001	1,028
Liberal Democrat	-1,552	261	[-2,063, -1,041]	<0.001	1,036
Liberal Democrat + notional margin	-1,554	261	[-2,067, -1,042]	<0.001	1,036

Table A38: Perceived party viability and the deprivation shift

Estimates per old constituency deprivation SD. Standard errors and 95% confidence intervals use clustering by old origin and new destination.

Outcome / sample	Estimate	SE	95% CI	p	N
Panel A. Change in perceived chance of winning the local constituency, W27 to W28 (0 to 100 scale)					
Reform/Brexit	-1.20	0.61	[-2.39, -0.01]	0.048	12,237
Conservative	-1.57	0.60	[-2.74, -0.39]	0.009	12,798
Labour	1.52	0.66	[0.22, 2.83]	0.022	12,470
Liberal Democrat	-0.12	0.55	[-1.20, 0.97]	0.835	12,676
Green	-0.08	0.52	[-1.09, 0.94]	0.881	12,608
Panel B. Effect on Reform support (pp), excluding strategic voters					
Baseline, full primary sample	2.51	0.87	[0.81, 4.20]	0.004	7,685
Excluding vote for lack of a local chance	2.31	0.83	[0.69, 3.94]	0.005	6,603
Excluding tactical voting website users	2.42	0.87	[0.71, 4.13]	0.006	7,461

Table A39: Effect on Reform support by incumbency

Group	Estimate	SE	95% CI	p	N
Same MP stood again	not estimable				
MP changed with the boundary review	2.07	1.77	[-1.41, 5.55]	0.243	729
Retired MP	1.78	2.27	[-2.66, 6.23]	0.432	1,303
Joint test of equal effects, $\chi^2(2) = 0.19$				0.909	7,685

Table A40: Leaflet content by deprivation of the new constituency, 2024

Open Elections human coded leaflets. Linear probability models estimated separately by party. Content estimates are percentage points per deprivation SD across the 543 English new constituencies. Standard errors are clustered by new constituency. Intervals are 95% confidence intervals. q uses Benjamini and Hochberg correction across the same 70 party by content tests in both displays. Share is the percentage of the party's leaflets carrying the code.

Code	Estimate	SE	95% CI	p	q	Share
Conservative ($N = 370$ leaflets, 168 constituencies)						
Brexit and Europe	2.5	2.0	[−1.4, 6.4]	0.209	0.457	8
Crime	−4.1	3.6	[−11.1, 3.0]	0.257	0.514	56
Economy	1.2	1.9	[−2.4, 4.9]	0.516	0.752	89
Education	1.2	3.1	[−5.0, 7.3]	0.708	0.901	61
Environment	−10.5	3.1	[−16.7, −4.4]	<0.001	0.007	60
Health	−7.1	4.2	[−15.2, 1.1]	0.089	0.291	69
Housing	−12.8	3.5	[−19.7, −5.9]	<0.001	0.003	44
Immigration	−0.6	3.2	[−6.8, 5.6]	0.846	0.955	37
Social welfare	−5.1	3.3	[−11.5, 1.3]	0.118	0.330	48
Transport	−5.0	3.6	[−12.0, 2.1]	0.166	0.410	61
Candidate	2.3	1.0	[0.3, 4.3]	0.023	0.136	95
Party leader	−2.4	3.2	[−8.6, 3.9]	0.457	0.718	27
Attack on other party	3.8	2.8	[−1.6, 9.3]	0.170	0.410	75
Tactical situation	3.6	3.5	[−3.2, 10.3]	0.302	0.541	34
Labour ($N = 408$ leaflets, 175 constituencies)						
Brexit and Europe	−0.2	0.7	[−1.6, 1.2]	0.758	0.910	2
Crime	3.9	3.3	[−2.5, 10.3]	0.234	0.481	39
Economy	4.3	2.1	[0.1, 8.5]	0.043	0.190	76
Education	4.9	2.3	[0.4, 9.3]	0.031	0.156	64
Environment	−12.3	2.8	[−17.7, −6.9]	<0.001	<0.001	38
Health	1.2	2.5	[−3.7, 6.1]	0.625	0.858	84
Housing	5.8	3.2	[−0.4, 12.1]	0.068	0.266	45
Immigration	13.4	2.8	[7.8, 19.0]	<0.001	<0.001	38
Social welfare	−1.0	1.5	[−3.9, 1.9]	0.488	0.727	9
Transport	0.1	3.0	[−5.9, 6.0]	0.983	0.995	23
Candidate	3.8	1.5	[0.7, 6.8]	0.015	0.115	90
Party leader	8.8	2.4	[4.1, 13.6]	<0.001	0.003	29
Attack on other party	5.1	2.9	[−0.5, 10.8]	0.076	0.281	68
Tactical situation	−16.1	2.4	[−20.8, −11.3]	<0.001	<0.001	26
Liberal Democrat ($N = 415$ leaflets, 158 constituencies)						
Brexit and Europe	8.1	3.7	[0.7, 15.4]	0.031	0.156	8
Crime	−4.9	3.4	[−11.6, 1.8]	0.151	0.407	20
Economy	3.2	3.0	[−2.7, 9.2]	0.288	0.535	77
Education	0.1	3.6	[−6.9, 7.1]	0.986	0.995	22
Environment	5.5	2.8	[−0.0, 10.9]	0.051	0.208	81
Health	4.7	2.2	[0.3, 9.1]	0.036	0.167	88
Housing	−1.2	4.7	[−10.4, 7.9]	0.793	0.910	31
Immigration	−0.1	1.5	[−3.0, 2.7]	0.920	0.990	3
Social welfare	4.2	4.3	[−4.3, 12.7]	0.330	0.577	12
Transport	−3.0	4.1	[−11.1, 5.1]	0.471	0.718	23
Candidate	3.4	1.4	[0.6, 6.2]	0.018	0.119	95
Party leader	1.7	2.4	[−3.0, 6.5]	0.472	0.718	12
Attack on other party	−1.3	3.4	[−7.9, 5.3]	0.700	0.901	81
Tactical situation	−25.5	4.4	[−34.2, −16.9]	<0.001	<0.001	71

Table A41: Leaflet content and volume by deprivation of the new constituency, 2024

Open Elections human coded leaflets. Linear probability models estimated separately by party.
Content estimates are percentage points per deprivation SD across the 543 English new constituencies.
Standard errors are clustered by new constituency. Intervals are 95% confidence intervals.
q uses Benjamini and Hochberg correction across the same 70 party by content tests in both displays.
Share is the percentage of the party's leaflets carrying the code.

Code	Estimate	SE	95% CI	p	q	Share
Green ($N = 165$ leaflets, 103 constituencies)						
Brexit and Europe	−1.3	0.9	[−3.2, 0.5]	0.168	0.410	1
Crime	2.2	2.4	[−2.6, 6.9]	0.370	0.608	7
Economy	−0.3	6.3	[−12.6, 12.0]	0.962	0.995	48
Education	3.1	5.1	[−6.9, 13.1]	0.543	0.760	19
Environment	1.2	4.5	[−7.7, 10.1]	0.785	0.910	79
Health	4.2	3.3	[−2.3, 10.7]	0.203	0.457	80
Housing	6.3	5.7	[−5.0, 17.5]	0.273	0.531	58
Immigration	−1.3	3.5	[−8.2, 5.5]	0.699	0.901	9
Social welfare	7.0	4.1	[−1.1, 15.1]	0.089	0.291	7
Transport	0.1	5.1	[−9.9, 10.1]	0.979	0.995	24
Candidate	3.4	2.1	[−0.7, 7.4]	0.107	0.312	92
Party leader	−8.9	2.9	[−14.5, −3.2]	0.002	0.019	13
Attack on other party	9.3	4.0	[1.6, 17.0]	0.019	0.119	75
Tactical situation	−7.4	5.8	[−18.8, 3.9]	0.200	0.457	52
Reform ($N = 185$ leaflets, 139 constituencies)						
Brexit and Europe	2.5	4.0	[−5.3, 10.2]	0.537	0.760	68
Crime	1.3	3.3	[−5.2, 7.7]	0.703	0.901	14
Economy	−0.3	2.7	[−5.6, 5.0]	0.915	0.990	93
Education	−3.3	3.2	[−9.5, 2.8]	0.290	0.535	16
Environment	−5.3	3.2	[−11.6, 1.0]	0.101	0.306	21
Health	0.0	3.0	[−5.9, 5.9]	0.995	0.995	84
Housing	2.5	2.8	[−3.0, 8.1]	0.373	0.608	14
Immigration	−3.7	2.2	[−8.1, 0.6]	0.091	0.291	95
Social welfare	1.5	1.5	[−1.5, 4.5]	0.338	0.577	4
Transport	−3.2	2.7	[−8.4, 2.1]	0.233	0.481	6
Candidate	−0.5	1.6	[−3.6, 2.5]	0.737	0.910	98
Party leader	1.3	4.4	[−7.3, 9.9]	0.763	0.910	63
Attack on other party	0.7	4.1	[−7.3, 8.7]	0.862	0.958	19
Tactical situation	−0.4	1.3	[−3.0, 2.2]	0.784	0.910	3

Leaflet volume by party. HC1 standard errors and 95% confidence intervals.
Counts cover 543 constituencies. Shares cover 306 constituencies with a recorded leaflet.
Estimates are per deprivation SD across English new constituencies. Counts are leaflets per constituency; shares are percentage points.
Final column gives mean count or mean percentage share. Blank q cells are outside the content correction family.

Conservative, leaflets per constituency	−0.29	0.05	[−0.39, −0.18]	<0.001		0.68
Labour, leaflets per constituency	−0.16	0.06	[−0.28, −0.04]	0.007		0.75
Liberal Democrat, leaflets per constituency	−0.49	0.08	[−0.66, −0.33]	<0.001		0.76
Green, leaflets per constituency	−0.15	0.04	[−0.23, −0.06]	<0.001		0.30
Reform, leaflets per constituency	−0.09	0.02	[−0.14, −0.04]	<0.001		0.34
Conservative, share of constituency leaflets (pp)	−3.5	1.6	[−6.7, −0.4]	0.029		23.2
Labour, share of constituency leaflets (pp)	7.5	2.0	[3.6, 11.4]	<0.001		25.0
Liberal Democrat, share of constituency leaflets (pp)	−6.8	2.2	[−11.1, −2.4]	0.002		23.9
Green, share of constituency leaflets (pp)	−0.1	1.4	[−2.9, 2.7]	0.938		11.1
Reform, share of constituency leaflets (pp)	2.9	2.0	[−1.0, 6.9]	0.148		16.8

Table A42: Issue ownership of immigration and the deprivation dose

Party best on MII	W25	W26	W27	W28	W29
Panel A. Best party on the most important issue (MII), respondents whose MII is immigration (weighted %)					
Con	21.4	15.7	15.8	11.8	10.5
Lab	1.9	3.4	3.3	2.2	3.0
LD	0.6	0.4	0.2	0.1	0.2
Green	0.2	0.2	0.1	0.1	0.1
Reform	25.6	33.0	41.3	54.1	51.4
Other	3.8	1.6	1.3	0.8	1.0
None/DK	46.5	45.8	38.1	30.8	33.9
Respondents (N)	3,243	6,069	5,587	6,225	5,818

Party could reduce immigration	W27	W28	W29
Panel B. Party could reduce immigration (weighted %, yes among valid responses)			
Reform	35.5	38.9	43.8
Con	20.0	13.6	17.7
Lab	14.7	12.7	15.2
LD	4.0	2.9	3.5
None	41.5	43.7	35.1
Respondents (N)	21,213	23,823	5,517

Estimate	Coefficient	SE	95% CI	N
Panel C. Dose effects, W26 to W28				
Immigration salience	−0.0140	0.0094	[−0.0326, 0.0045]	7,780
Reform best on MII	0.0056	0.0101	[−0.0141, 0.0253]	5,768
Panel D (exploratory). Heterogeneity by pre-treatment W25 ownership				
Immigration MII (W25) ($N = 4,818$; $n = 725$ where moderator = 1, $n = 4,093$ where moderator = 0)				
Dose × moderator	0.0058	0.0332	[−0.0594, 0.0709]	4,818
Dose effect, moderator = 0	0.0267	0.0128	[0.0015, 0.0519]	4,818
Dose effect, moderator = 1	0.0325	0.0326	[−0.0314, 0.0963]	4,818
Reform best on MII (W25) ($N = 4,818$; $n = 278$ where moderator = 1, $n = 4,540$ where moderator = 0)				
Dose × moderator	0.0349	0.0360	[−0.0356, 0.1053]	4,818
Dose effect, moderator = 0	0.0249	0.0129	[−0.0004, 0.0501]	4,818
Dose effect, moderator = 1	0.0597	0.0353	[−0.0095, 0.1289]	4,818
Reform best and immigration MII (W25) ($N = 4,818$; $n = 211$ where moderator = 1, $n = 4,607$ where moderator = 0)				
Dose × moderator	0.0478	0.0442	[−0.0389, 0.1345]	4,818
Dose effect, moderator = 0	0.0256	0.0128	[0.0005, 0.0506]	4,818
Dose effect, moderator = 1	0.0734	0.0439	[−0.0127, 0.1594]	4,818

Table A43: Deprivation shift and the change in Reform support

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Deprivation shift (1 SD)	2.51	2.34	2.66	2.35	2.76	2.48	2.18
	(0.87) [0.81, 4.20]	(0.79) [0.80, 3.88]	(0.78) [1.12, 4.20]	(0.85) [0.69, 4.01]	(0.88) [1.03, 4.48]	(0.90) [0.72, 4.24]	(0.80) [0.62, 3.75]
Endpoint	W29	W28	Mean of W28 and W29	W29	W29	W29	W29
Deprivation measure	Income	Income	Income	Income	Employment	IMD	Income
Predetermined covariates	No	No	No	Yes	No	No	No
Observed geography	Stable observed W19 to W29						Endpoint stability
Old constituency fixed effects				Yes			
Clustering	Two way old and new constituency						
Respondents	7,685	8,837	7,685	7,685	7,685	7,685	8,766

Table A44: Measured proxy and exploratory response diagnostics

Account tested	Estimate	95% interval
Panel A. Confirmatory tests		
Perceived neglect, politicians do not care, W26 to W27	+0.005	$[-0.034, +0.043]$
Perceived neglect, trust in MPs, W26 to W29	−0.004	$[-0.064, +0.055]$
Attention to rhetoric, interaction with dose	−0.07	$[-1.75, +1.60]$
Panel B. Exploratory tests		
Con+LD spending experienced (per SD)	−0.25	$[-1.74, +1.24]$
Conservative recapture from W26 Reform or undecided origins	+3.15	$[-0.26, +6.55]$
Immigration priority, interaction with dose (W25 measure, common sample)	+0.52	$[-6.01, +7.05]$
Latent Reform support among available non intenders (pooled)	+4.24	$[-2.85, +11.33]$
Strategic voting, excluding voters whose favourite had no chance	+2.31	$[+0.69, +3.94]$

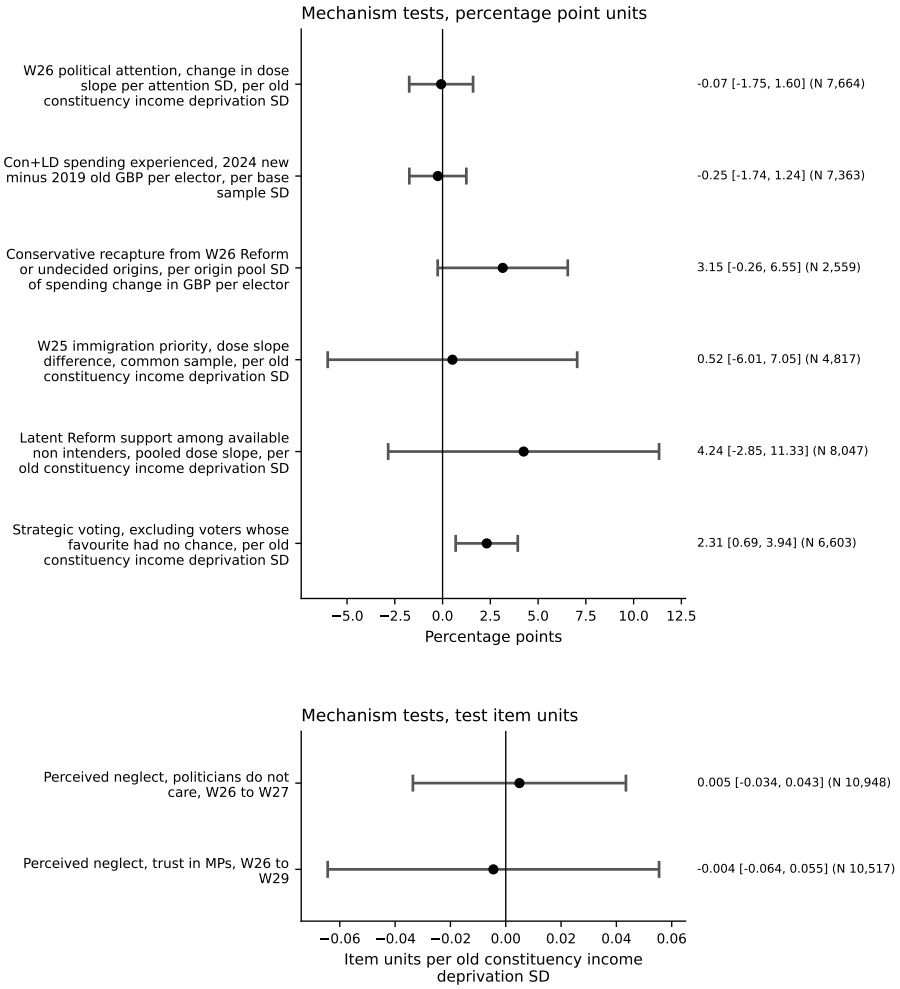


Figure A21: Diagnostics of constituency context and Reform support

Table A45: Design-strengthening tests

Specification	Estimate (pp)	SE	95% CI	p	N
Panel A. Placebo elections.					
2015 placebo (RR, W5 to W6)	-1.00	1.34	[-3.63, 1.63]	0.457	1,085
2017 placebo (RR, W12 to W13)	-0.53	1.54	[-3.55, 2.49]	0.729	1,915
2019 placebo (RR, W18 to W19)	-0.81	0.93	[-2.63, 1.01]	0.383	2,918
2024 actual reform (RR, W26 to W29)	2.51	0.87	[0.81, 4.20]	0.004	7,685
Panel B. Pooled post-period targets.					
W29 only	2.51	0.87	[0.81, 4.20]	0.004	7,685
Mean W28 to W29	2.66	0.78	[1.12, 4.20]	<0.001	7,685
Mean W27 to W30	2.13	0.74	[0.69, 3.58]	0.004	7,685
Panel C. Lagged-outcome bracketing (W29).					
First difference, W26 to W29	2.51	0.87	[0.81, 4.20]	0.004	7,685
Lagged outcome + W19 to W25 lags, W29	2.71	0.89	[0.97, 4.45]	0.002	7,685
Panel D. Aggregate overlap and destination deprivation.					
Aggregate overlap relative to old seat	1.42	1.18	[-0.89, 3.73]	0.228	7,663
New seat rel. old seat	1.41	1.28	[-1.10, 3.91]	0.272	7,663

Table A46: Identification diagnostics

Diagnostic	Wald statistic	df	p	N
Reform outcome leads jointly zero	2.16	6	0.905	7,685
Predetermined covariates jointly balanced	46.14	18	<0.001	2,435
Local service trajectory leads jointly zero	12.79	11	0.307	6,806
Area battery jointly balanced	25.54	5	<0.001	583

Table A47: Entropy balancing estimates for the deprivation dose effect on Reform support

Reform support change, W26 to W29, percentage points per old constituency deprivation SD (5.31 deprivation percentage points).

Old origin fixed effects. Standard errors and 95% confidence intervals use clustering by old origin and new destination.

Entropy convergence and weight diagnostics, main coding and Don't know kept coding respectively

Diagnostic	Main coding	Don't know kept coding
Iterations	5	5
Kish effective N	2395.1	2604.3
Maximum weight	2.24	2.25
Maximum absolute weighted dose covariate covariance	2.4e-11	4.4e-11

Balance is imposed by weighting.

Specification	Coef.	SE	95% CI	p	N
Full sample, main coding, unweighted	2.51	0.87	[0.81, 4.20]	0.004	7,685
Full sample, Don't know kept coding, unweighted	2.70	0.95	[0.85, 4.56]	0.004	9,207
Complete covariate sample, main coding, unweighted	4.27	1.57	[1.19, 7.34]	0.007	2,435
Complete covariate sample, main coding, entropy weighted	5.30	1.99	[1.39, 9.21]	0.008	2,435
Complete covariate sample, Don't know kept coding, unweighted	3.93	1.50	[0.99, 6.88]	0.009	2,648
Complete covariate sample, Don't know kept coding, entropy weighted	4.60	1.98	[0.72, 8.49]	0.020	2,648

Table A48: Covariate balance after entropy weighting, standardised dose associations

Complete W25 covariate samples after entropy weighting. Standardized covariate slopes per old constituency deprivation SD.

Main coding N 2,435. Don't know kept coding N 2,648. Balance is imposed by weighting.

Covariate units are standard deviations. Covariance estimation uses clustering by old origin and new destination.

Covariate	Main coding	Don't know kept coding
Age (W25)	-0.002	-0.002
Female	0.004	0.004
Degree qualification (W25)	0.001	0.001
No qualifications (W25)	-0.002	-0.003
Household income band (W25)	-0.004	-0.004
Personal income band (W25)	0.000	0.000
Employed (W25)	0.000	0.000
Unemployed (W25)	-0.001	-0.001
Homeowner (W25)	-0.002	-0.003
Social renter (W25)	0.003	0.003
Nonwhite ethnicity (W25)	0.000	0.001
Left to right self placement (W25)	0.001	0.001
Reform party identification (W25)	-0.001	0.000
Reform vote intention (W25)	-0.002	-0.001
Recorded EU referendum Leave vote	0.001	0.002
Recorded Conservative vote (2019)	-0.002	-0.001
Recorded Labour vote (2019)	0.000	-0.001
Recorded Reform/Brexit vote (2019)	0.001	0.002

Table A49: Joint covariate balance tests before and after entropy weighting

Joint Wald tests of 18 predetermined W25 covariates in complete covariate samples.

Covariance uses clustering by old origin and new destination. After weighting, balance is imposed by construction.

Specification	Wald statistic	df	p	N
Main coding, before weighting	46.14	18	< 0.001	2,435
Main coding, after weighting	0.03	18	1.000	2,435
Don't know kept coding, before weighting	57.69	18	< 0.001	2,648
Don't know kept coding, after weighting	0.05	18	1.000	2,648